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A model specification for FRP composites for civil engineering structures

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Abstract

A proposed model specification for FRP composite materials for use in civil engineering structural systems is described in this article. The model specification provides a classification systems for FRP materials, describes admissible constituent materials and limits on selected constituent volumes, describes tests for specified mechanical and physical properties, specifies limiting values of selected properties in the as-received condition and in a saturated state, and provides a protocol for predicting long-term property values subjected to accelerated aging based on the Arrhenius model. The model specification is included as an appendix to the article.

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1. Introduction

It is widely recognized that in order for fiber reinforced polymer (FRP) composite materials to be used in the construction of civil engineering structures such as buildings and bridges a uniform procedure for specifying these materials is required. Standard specifications exist for all commonly used materials in the civil engineering construction. These specifications ensure that materials used in civil engineering projects are defined in specific classes, are tested using standard procedures, are certified in a uniform format and provide specific properties for their intended use. A consensus-based general material specification for FRP materials for use in civil engineering structural applications does not exist at this time. A model specification has been developed by the authors, under sponsorship of the US Federal Highway Administration (FHWA) and in coordination with the American Association of State Transportation and Highway Officials (AASHTO). The specification has not yet been approved by either AASHTO or the American Society for Testing and Materials (ASTM). This article describes the development of the model FRP material specification and the key elements

that the specification contains. The Appendix to this article contains the model specification itself. The specification is titled ‘Standard Specification for Fiber Reinforced Polymer (FRP) Composite Materials for Highway Bridge Applications’ as per the requirements of the contract under which it was developed.

The model specification was developed by a team of researchers who have extensive experience and expertise in the use of FRP materials for civil engineering structures and extensive prior expertise in the development of material specifications. The model specification was developed in the following steps: (a) technical literature on the subject of characterization of the mechanical and physical properties of FRP composite materials for both short-term and long-term properties was studied from the perspective of writing a specification; (b) existing material and design codes and specifications for composite materials were reviewed and evaluated; (c) existing design codes for conventional materials were reviewed to determine their relationship to material specifications; (d) key elements for a FRP material specification for civil engineering structures were identified in consultation with design professionals and end-users, state and federal officials; and (e) draft specifications and commentaries were developed at 30, 60 and 90% completion targets for review.

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From a detailed review of codes and specifications for composite materials a number of key sources were identified as a basis for the development of the model FRP material specification for civil engineering applications. These documents, detailed below, provide procedures for material characterization, methods for prediction of long-term properties and performance, and acceptance criteria. The American National Standard for Ladders [1], the specification for reinforced plastic ladders, provides detailed procedures for testing and minimum properties for acceptance of FRP materials for use in ladders. Tests for physical properties (e.g. density, maximum water absorption and cure) and mechanical properties subjected to dry, wet, elevated temperature and weathered conditions are stipulated. The International Conference of Building Officials (ICBO) Acceptance Criteria AC-125 [2] specifies selected physical and mechanical properties to be measured and reported for composite materials used for repair and retrofit of concrete structures. While no minimum properties are specified for use, limits on minimum property retention values following conditioning for 1000 and 3000 h are stipulated. The US Department of Defense Military Handbook 17 [3] provides procedures for obtaining properties for design for FRP composites for aerospace applications, as well as property data for specific composite material systems. Finally, specifications of the American Society of Testing and Materials (ASTM) related to ‘fiberglass’ tanks, pipes and poles (e.g. ASTM D2997, D3754, D4021 and D4923) provide guidance on test methods, acceptance criteria and methods for prediction of long-term properties of FRP composites [4].

Key sections of the specification are discussed in the text that follows. The order of the discussion follows that of the specification, which is organized and presented in the generally-accepted format provided by ASTM [5]: scope, classification, materials, manufacturing, qualification testing, acceptance testing, reporting and quality assurance. Sections on terminology, ordering information, keywords and product marking are contained in the specification but are not discussed in the article. References to tables, figures and text sections that are numbered with the decimal point (e.g. Section 9.5.2) refer to elements of the specification and not to the article itself.

2. Scope

According to ASTM, a specification is an explicit set of requirements to be satisfied by a material, product or system. A material specification serves three main purposes: (1) to aid in the completion of purchasing agreements between materials suppliers and purchasers, so that all batches and lots of a material conform to the requirements; (2) to define standard classes and forms

of the material; and (3) to identify performance data that must be disclosed as part of the material purchase [5].

In addition, the FRP materials specification was developed to apply to a wide range of FRP composite materials for numerous different uses, while still ensuring quality and promoting durability. The specification focuses on materials most likely to meet AASHTO’s stated goals of providing a 75-year life in its structures. It was determined that the specification should, at a minimum: (1) classify FRP materials into groups so that similar materials will be tested in an identical manner and will meet the same minimum performance requirements; (2) require that material manufacturers provide sufficient property data for structural design using the FRP materials; (3) ensure that high quality constituent materials and well-controlled manufacturing processes are used to produce the FRP materials; (4) provide long-term data on mechanical property retention and a method for service life prediction; and (5) provide quality assurance procedures so that agencies procuring the material can verify that FRP materials meet the specification.

FRP composite parts covered by the specification are made of one or more ‘qualified’ laminates. The qualifying procedure includes a number of mechanical and physical screening tests. In the specification itself and in this article, the term qualification implies a set of tests that are completed on trial laminates or on laminates cut from production parts. The qualification tests are Procedure A, which provides a wide range of test data and screens for key properties, and Procedure B, which provides long-term property retention data on the material. The parts themselves are ‘accepted’ if the testing completed on coupons cut from production parts shows that the material properties are essentially equivalent to those of the qualified laminates. The testing regime for part acceptance is a small subset of the tests required for laminate qualification. The acceptance tests are Procedure C, which provides a comparison test to show that the material being accepted is substantially the same as the qualified (Procedure A) material, and Procedure D, which requires that the material retain key mechanical and physical properties in a hot–wet environment. The step-by-step procedure for qualification testing and subsequent acceptance testing is described in Fig. 1.

It is important to note that the materials specification only covers coupon-level properties. In some applications a materials specification alone will be sufficient to specify an FRP structural element. In many cases additional element-level specifications that consider full-sectional mechanical behavior, bond and anchorage properties, and/or connections may be necessary. For complex FRP parts, full-section behavior will be even more difficult to predict from coupon data. The effect

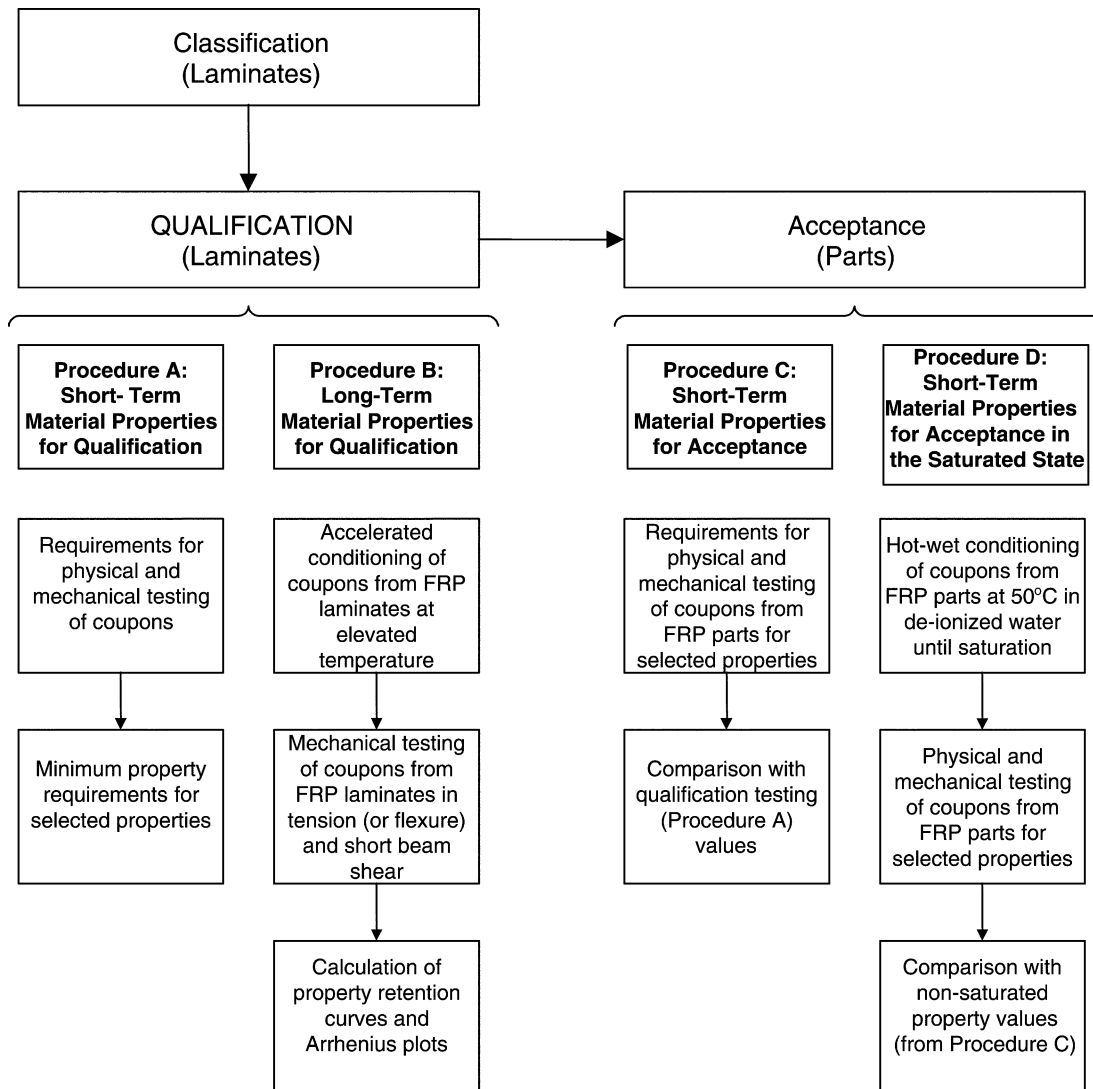


Fig. 1. Flowchart of required qualification and acceptance testing.

of T and L junctions, thickened regions and ply drops simply cannot be predicted at the coupon level. Furthermore, with the exception of laminate thickness, the geometric tolerances of the FRP part, such as straightness and twist, are not considered, as these cannot be ascertained at a coupon scale. Finally, the materials specification requires that a wide range of mechanical property data be collected and published, but it does not provide design allowable stresses or strength-reduction factors for design. These design specifications are still being developed for FRP composites, and design specifications for FRP composite concrete reinforcements and for FRP composite highway sign supports have recently been published [6,7].

3. Classification

Laminates supplied according to this specification are classified according to fiber volume fraction, percentage

of fiber oriented in the longitudinal direction, fiber type and resin type. The purpose of the classification system is to provide broad categories of FRP materials, so that minimum properties for each of the broad categories can be specified. The classification of materials takes place on the laminate level. In the general case, it is not possible to classify a complex FRP part itself, because such a part could be constructed of multiple different laminates. Annex B2 of the specification presents such an example. Therefore, each laminate within the part is classified and tested.

A laminate is considered to be a relatively thin plate, which has two dimensions that are considerably larger than the third (thickness) dimension. A laminated composite is generally envisioned as being made of discreet plies or laminae with identifiable fiber orientation and properties in each ply. This draft specification does not deal with composites on the ply level, and in fact

anticipates that many of the composites that meet the specification will not be ‘laminated’ per se. In many plates that are reinforced primarily with rovings or tows, no laminated structure is identifiable. However, since three-dimensional reinforcements are rarely used, the laminated assumptions apply. A laminated structure is much more evident in plates constructed with woven rovings, non-woven fabrics or stitchmat, for example. According to this specification, small regular shapes such as round rods, square bars and narrow strips can also be considered to be laminates. Therefore, as it is used in this specification, a laminate is the basic building block of a composite part.

The primary resin systems currently in use in structural FRPs are included in the specification. The specification does not include thermoplastic polymers or phenolics at this time, because these resin systems are not in widespread use in structural applications, and because the mechanical properties and durability of FRP composites made with these resins has not been provided in or demonstrated by documented laboratory research and field application.

The specification allows for the use of glass and carbon fibers. The sub-type of glass or carbon fiber is not limited. Both E-glass and S-glass and both PAN-based and pitch-based carbon fibers are permitted. Aramid fibers are not included at this time due to the small number of aramid-fiber FRPs used in infrastructure applications and due to the lack of laboratory and field data on these FRPs. Hybrid FRP composites using a mixture of glass and carbon fibers are permitted as long as the secondary fiber volume is less than 20%. This allows for the specification to use the same mechanical property requirements for a given class of FRP and for hybrids based on that class. So, for example, the same minimum property requirements hold for a carbon fiber type 1 laminate and any hybrid based on that class.

The primary indicator of mechanical performance in an FRP composite is the volume or weight fraction of fiber reinforcement in the composite. It is the fiber in an FRP composite that primarily provides the desired strength and stiffness of the composite material. The fiber content should, therefore, be as high as reasonably possible within the FRP composite. This fiber content can be expressed as a volume fraction or as a weight fraction. In this specification, volume fractions for fibers are always used, because weight fractions cannot be compared when dealing with fibers of different densities (glass and carbon, for example). The specification uses this primary characteristic of an FRP composite—the volume fraction of fiber—as a means of classifying the composite. The theoretical limit of fibers in a unidirectional composite approaches 90% [8]. Practical limits for unidirectional composites are approximately 60%, with limits for composites with transverse reinforcements dropping to 50% and below depending on the

level of transverse reinforcement and the type of fabrics used.

In this specification, three lower bounds or ‘minimum levels’ of fiber volume are given. These levels are 30, 40 and 50%, depending on classification. No upper bound of fiber reinforcement is provided, and it is anticipated that many FRP composites produced to meet this specification will have fiber volume fractions that exceed the lower bounds. The lower bound is provided to ensure that only high performance materials are admitted under this specification. Materials with lower volume fractions should be considered non-structural, as their strength and stiffness will dictate that they must be used at low sustained stress levels. Low volume fraction FRP materials may have applications in bridge structures for non-sustained load applications, such as in the rehabilitation of cracked masonry walls in a bridge pier.

In addition to specifying a minimum fiber volume fraction, the specification also provides limits on the orientation of this fiber within the part. Fiber orientation is generally expressed as an angle relative to the longitudinal direction of the part (e.g. 0° fiber, 90° fiber). In most structural elements, the normal stresses along the major axis of the part are the principal design driver, and, therefore, the fiber reinforcements within the FRP composite are oriented primarily to resist these stresses. The fibers oriented along this longitudinal axis are typically described as the longitudinal fibers or 0° fibers. In FRP composites used as concrete reinforcements (rods, tendons), almost all of the fiber will be longitudinal fiber. For laminates that are resisting biaxial forces or shear, the fibers will be distributed between the longitudinal and other directions (90°, $\pm 45^\circ$, etc). Many parts also contain layers of fiber made of continuous or chopped strand mat, which have fibers that run in a random, swirled pattern and are thus omni-directional. Table 1 describes the types of laminates defined by the specification and examples of FRP material parts that generally fall into each of these types.

The type 4 laminate is included in the specification to allow for the use of relatively low volume FRP composites as field-placed and field-cured reinforcements for existing structures. Shop-produced laminates that are consolidated using manual methods or open molded may also have low volume fractions and may be admitted if they meet type 4 volume fraction requirements.

4. Materials

The specification admits a wide range of constituent materials: resins, fibers, cure systems, veils, etc. Furthermore, the specification does not require the testing of constituent materials, so that the testing of fiber reinforcements or of neat resins is not required. In most

Table 1
Laminate types and example applications

Laminate type	Minimum fiber volume fraction (v_f/v_c) (%)	Minimum longitudinal fiber ratio (v_{long}/v_f) (%)	Representative FRP part or structural element
1	50	95	tendon, rebar, dowel bar, strengthening strip
2	40	75	profile shape, molded profile
3	40	40	molded plate, deck
4	30	95	handmade strip, deck, field wrap

instances, the specification does not attempt to control or limit the types and amounts of constituent materials. Rather, the specification ensures that a quality final product is produced through performance testing of the final product in qualification and acceptance tests as described in Sections A.8 and A.9 of the specification. In cases where the quality of the FRP composite may be compromised by the use of a specific component or by the use of too much of a component, then the use or quantity of that component is restricted.

At this time, the specification covers only isophthalic polyester, vinylester and epoxy resin. The use of isophthalic resins is limited to applications that will not put the FRP into contact with hydraulic cement concrete. The concrete porewater environment is known to be highly alkaline and thus can corrode glass fibers. A number of researchers in this field have concluded that polyester resins are not appropriate for use in concrete reinforcements or in concrete repair materials [9–11]. In addition, the glass transition temperatures of polyester resins are generally lower than those of vinylester resins. Moisture uptake in polyester resins is also generally higher than in vinylester resins [12,13]. The specification permits the use of carbon or glass fibers. At this time, aramid fiber FRP composites are not included in the specification. The use of aramid fibers is limited to a few products at this time. Furthermore, the durability of aramid FRPs in moist environments has been questioned by some durability researchers [14].

Inert fillers are used to dilute the resin system to improve processability, improve specific physical properties (such as fire resistance) and to decrease cost. High levels of added filler in resins may reduce durability. Consequently, the quantity of filler is limited to 20% by weight of the base resin system. The quantity of additives in the resin system is given by weight as this is the method used in current practice to measure out constituent materials for processing. The quantity of fiber used to predict properties and to classify the material, on the other hand, is given by volume. Additives are not limited except in the case of low-profile thermoplastic shrink additives, which may lower the glass transition temperature of the matrix and thus may affect the long-term performance of the FRP composite.

5. Manufacturing

The specification allows any method of production for the composite material as long as a fully-cured composite is produced. Research has shown that residual monomer left in the polymer can lead to problems with long-term durability [13]. Furthermore, the glass transition temperature of the polymer increases with the degree-of-cure [15]. Therefore, it is important that a full cure should be achieved. Physical testing described in Qualification and Acceptance Testing, below, is used to illustrate that a sufficient cure has been achieved in the FRP composite.

Because the specification covers materials produced using any means, and also covers materials that are measured in lineal measurements (such as concrete reinforcing bars) or in areal measurements (such as pultruded plates), a standard means for determining the size of a lot of material cannot be provided by the specification. In general, a new lot starts whenever a new batch of constituent materials is used, or when a stop-and-start process such as closed molding finishes. For processes that are essentially continuous, a lot may be defined as the amount of material produced in one shift on the production line. The specification requires that the manufacturer provide a definition of lot size. Quality assurance provisions of the specification use this definition for sampling and testing.

It is envisioned that the manufacturer will qualify a laminate according to the provisions of Section A.8 of the specification and then use that laminate in production parts, with the laminates being cut from the production parts undergoing acceptance testing as described in Section A.9. If any substantive aspect of the constituent materials change, or if any substantive aspect of the production process changes, then the laminates being used in the parts can no longer be considered 'qualified'. In this case, new laminates must be qualified for the production part and coupons cut from the new production part must be re-accepted. Substantive changes include, for example, a change in the resin supplier or the amount of filler use in the matrix. It is possible that some adjusting of production parameters, for example, an increase or decrease in the level of catalyst to account

for changes in ambient temperature and relative humidity, will be necessary during processing. Most changes of this nature are considered to be part of process control, and thus need not necessarily be considered as producing a 'new' laminate that needs to be qualified and accepted.

6. Testing for laminate qualification

The testing specified in Section A.8 accomplishes three objectives: (1) providing materials data for preliminary design to the purchaser; (2) demonstrating material quality and suitability by showing that key material properties satisfy limiting values; and (3) providing accelerated test data to allow for the prediction of long-term service lives. The set of mechanical and physico-chemical tests on the qualifying laminate are denoted as Procedure A in the specification. The long-term test procedure is denoted as Procedure B in the specification. The specific testing, calculations and comparisons required in Procedures A and B were outlined in Fig. 1.

6.1. Qualification laminates: identification and production

A laminate identification sequence is specified so that a common means of identifying materials may be used throughout the specification. A unique laminate is described by the classification type (which includes information regarding the fiber volume fraction and fiber orientation, fiber type and resin type of the laminate), by the thickness of the laminate, and by an alpha-numeric identifier that is assigned by the laminate manufacturer. This alpha-numeric identifier is an internal code assigned to differentiate amongst different products produced by the same manufacturer. This identifier is the manufacturer's responsibility, and is not explicitly dictated in the specification.

A special form of the qualification laminate is the test property laminate or TPL. The test property laminate is constructed and manufactured to represent the behavior of the laminate as if it were cut from the production part. The advantages of the TPL are: (1) that material development and qualification can take place on a representative element before the full-scale part is developed and without the expense of producing full-scale parts specifically for materials qualification; (2) transverse tests can be completed on coupons cut from TPLs that are made at the appropriate width; and (3) a given laminate that is qualified from a TPL can be used in many different FRP parts. It is required that the TPL be produced using the same resins, fibers, fabrics, stacking sequence and veils that will be used for the part itself. Furthermore, the thickness of the TPL must be the same as the thickness of coupons cut from the production part.

For compact shapes like solid round and square shapes such as those used for concrete reinforcing bars, it may be useful to use full-size shapes produced without surfacing treatments as a TPL. This is acceptable as long as the thickness dimension or dimensions of the TPL are the same as the average thickness of the production parts having some deformed geometry. In all the cases, it is acceptable to perform qualification tests on coupons from the FRP part and not from test property laminates.

6.2. Procedure A—short-term material properties for qualification

The primary purpose of the testing requirements in Procedure A is to ensure that extensive material property data are available to material purchasers and their structural designers. The properties required by the specification are no more extensive than the range of properties currently being reported by the major US suppliers of FRP composites. See, for example, the Extren Design Manual [16] and the Pultex Design Manual [17]. The specification only requires currently approved ASTM test methods for the characterization of FRP materials (see specification Table A.8.1). Where more than one test method is available for a certain property (for example, tension strength and modulus), the manufacturer may select the test method used to complete the tests from those given in the table. The specification does require only one test method to be used to produce a given test result, therefore, if tension test data is provided for Procedure A (qualification) testing via ASTM D 3039, then all other tension testing completed on the material as required in the specification (in Procedures B, C and D) must also be completed using D 3039. For essentially unidirectional (type 1) laminates, transverse testing is not required.

6.2.1. Mechanical property requirements

The material property limits given in Table A.8.2 of the specification are provided to ensure that the qualified materials have the minimum mechanical and physical properties that their fiber content and curing regime predict that they should have. Mechanical property limits are specific to each laminate type and fiber type, and were derived from test data reported on in the technical literature and from manufacturers' product literature (e.g. SikaWrap Hex 107G, Extren Design Manual, Pultex Global Design Guide, Dow Derakane 411-350). Cases of specific example laminates chosen at the 'boundaries' established in Section A4 of the specification were also examined using micro-mechanics. The limits given in Table A.8.2 are minimum requirements, which should be easily met by FRP laminates that are properly processed. The values contained in Table A.8.2 are minimum limits for the average results from the

specified tests. These values are not intended to be used as design strength values nor as lowest-common-denominator property values for manufacturers to aspire to. These are not the ‘strength-3 σ ’ values often used by design codes as the ‘guaranteed strengths’ of the materials. It is expected that many high quality FRP laminates will have mechanical properties much higher than those given in Table A.8.2, and the design codes will take advantage of these higher strengths in design equations developed for use in designing with these materials.

6.2.2. Physico-chemical property requirements

The minimum physico-chemical properties (given in Table A.8.2 of the specification) for glass transition temperature (T_g), Barcol hardness, fiber volume fraction and maximum moisture content are specified to ensure that the FRP composite matrix is manufactured properly. The glass transition temperature is an indirect measure of degree-of-cure in a thermosetting resin, and is an important indicator of material durability in an elevated temperature service environment.

The maximum moisture content gives data on the degree of cure of the laminate, on the void content of the laminate, on the proper bonding of the matrix to the fibers (the so-called interphase) and reflects on the probable durability of the composite in the civil engineering infrastructure environment [18,19]. A conditioning temperature of 122 °F (50 °C) has been selected for three reasons: (1) for thick laminates, it can take many months to reach equilibrium moisture content at room temperature and a moderately elevated temperature will dramatically decrease the time to reach equilibrium; (2) it is difficult to maintain a controlled room temperature without an expensive environmental chamber, whereas an inexpensive oven can be used to maintain 122 °F (50 °C); and (3) Procedure D conditioning in the hot wet environment is completed at 122 °F (50 °C), and thus moisture equilibrium data from Procedure A can be compared with mechanical test data gathered in Procedure D.

The Barcol hardness test is used to complement the measurement of glass transition temperature, as an indirect measure of the degree of cure of the FRP composite. The Barcol hardness test is inexpensive and quick, and may provide useful information on the distribution of cure throughout a part. The same test with the same lower limit of 50 is specified in the ANSI specification for reinforced plastic ladders [1].

The volume fraction requirements are the same as those given in Section A4 on laminate classification (see Table 1). A test for volume fraction is required to ensure that the laminate meets the total fiber volume fraction requirement as stated in the classification system. This testing is not required to demonstrate the distribution of fibers within the laminate. The laminate

stacking sequence is determined and controlled by the manufacturer.

6.2.3. Thermal analysis

The specification makes use of thermal analysis to determine the elevated temperature performance of the FRP composite materials. In general, thermal analysis testing and the enforced limits provided by the specification perform two functions: (1) ensure that the resin system selected for the part has sufficient elevated temperature performance for the infrastructure environment; and (2) ensure that the resin system has been processed in an appropriate way during product manufacture so as to bring the composite to a sufficient degree-of-cure. The term ‘fully-cured composite’ implies that 100% of the monomer present in the liquid resin system has been cross-linked into the solid thermosetting matrix as part of the curing process [3]. It is impossible to measure the degree-of-cure of a thermoset, but it is possible to infer that the composite is sufficiently cured. The approximate degree-of-cure can be established by comparing the T_g measured in the composite part with the T_g provided by the resin manufacturer. If the T_g in the part is well below that published by the manufacturer, then it can be concluded that the manufacturing process is not completely curing the part.

A large number of test methods and specialized equipment are available to measure the thermal performance of polymers. Each test gives a different measured outcome, but most are expressed as a temperature level at which some characteristic of the polymer changes. The most important temperature at which such a change takes place is known as the glass transition temperature (denoted T_g). In the simplest terms, the glass transition temperature is the temperature at which the polymer transitions from a rigid to a rubbery state and vice versa. Obviously, FRP composites produced for bridges are meant to perform in their rigid state, and it is, therefore, not acceptable to operate an FRP composite near, at, or above its glass transition temperature.

The difficulty arises because a number of test methods provide a glass transition temperature, but the temperatures they provide are not the same. Glass transitions in polymers occur over a range of temperatures, with an ‘onset’ temperature, a mid-point temperature and a ‘completed’ temperature. The change in state may be subtle and difficult to detect (it is not a change from a liquid to a gas for example, but rather a change from a rigid solid to a more rubbery solid).

Four primary methods have been considered as appropriate for measuring the glass transition temperature. These are differential scanning calorimetry (DSC), dynamic mechanical analysis (DMA), thermal mechanical analysis (TMA) and heat distortion temperature (HDT). The heat distortion temperature actually reports

a different thermal transition temperature that is proportional to, but generally lower than T_g [20].

These four types of testing are represented by many, more than four ASTM test methods. DSC compares temperature rise with the rate of heat uptake (defined as the heat capacity) to determine when the glass transition begins and ends. DMA compares the temperature rise with the vibrational (stiffness) characteristics of the composite to determine when the glass transition begins and ends. TMA compares the temperature rise with a change in the coefficients of thermal expansion (CTE) of the composite to determine when the transition occurs. Finally, HDT tests detect the onset of excessive deformation as a function of increasing temperature. HDT results are sometimes reported as heat distortion results and sometimes as temperature of deflection results (two different ASTM test methods) but the temperatures are often considered to be equivalent.

Only DSC testing does not depend on mechanical properties of the system. In the other test methods, the T_g measured is a function of the laminated structure of the composite and not just of the resin system itself. The same problem exists for any of the methods that measure thermal performance as a function of some mechanical parameter of the system. One solution to this problem is to cure separate resin—only coupons for thermal analysis testing (by leaving the fiber out). The danger here is that any thermal measurement made on specially-prepared coupons using resin only is unlikely to represent the state of cure in the FRP composite itself. Such a procedure might be useful in a research and development procedure, but cannot be used when the intent is to demonstrate the quality of the FRP composite that has been produced.

The strengths and weaknesses of each of these test methods are described in MIL Handbook 17, Volume 1, Section A6.4. MIL Handbook 17 recommends DMA as the method for determining the glass transition in composites [3]. However, most resin suppliers report the glass transition by DSC or report the heat distortion temperature of their resins. It is important that the composite manufacturer should be able to compare the glass transition of the processed laminate with the resin manufacturer's supplied data.

There do exist some resin systems where the glass transition is difficult to measure using DSC. Furthermore, in FRP materials with high fiber volume fractions, most of the mass being tested is fiber, and the transition in the polymer may be difficult to detect [3]. In these cases, the transition is essentially masked and cannot be detected by changes in the heat flow into the composite. Therefore, the specification does allow for the measurement of the glass transition using any of the methods listed in Table A.8.1 of the specification. Methods that provide a direct measure of T_g are preferred, and the specification notes this preference. The same tempera-

ture limits apply regardless of the method adopted. This may encourage material suppliers to adopt DSC testing as the standard, because for a given composite, the transition temperature as measured by DSC is usually higher than the temperature as measured by HDT [20]. The literature does not give a definitive picture of this differential and the specification cannot, therefore, provide a separate target for each of the acceptable test methods. The limits set on T_g were chosen based on two factors. First, manufacturers' literature was examined to determine the transition temperature expected for fully-cured resins. Second, the expected service temperature of the materials in highway bridge applications was taken into account. The interaction of these two factors were then qualified based on the glass transition temperatures reported in product literature of current composite manufacturers.

6.3. Procedure B—long-term material properties for qualification

The accelerated testing requirements outlined in Section A8.6 of the specification, and referred to as Procedure B testing are intended to provide the purchaser with long-term test data for the material under service conditions. Accelerated testing and mathematical modeling outlined in the specification are based on the Arrhenius relationship, which states that a chemical process is accelerated as an exponential function of temperature [21]. By conditioning and testing the materials at a number of different temperatures, the time rate of degradation at an average service temperature (considered to be 73 °F (23 °C) in the generic case) can be predicted. If the rate of change in mechanical properties in the service environment is known, then the service life of the material can be predicted. The application of Arrhenius modeling to civil engineering materials was first described by Litherland [22]. A recent application of the model to the degradation of polymeric materials used in civil engineering structures was presented by Iskander and Hassan [23].

The application of the Arrhenius model, with its inherent assumptions, to the change in mechanical properties in an FRP composite over time, is not without difficulty. The following assumptions are implicit. First, one chemical degradation mode must dominate in the change of material properties over time; the effects of mixed modes or changes in modes over time cannot be accommodated. Second, the conditioning of the material at elevated temperatures must not change the mode by which the material degrades under service temperatures. For this reason, elevated temperatures that approach the glass transition temperature of the composite are believed to be too high. Finally, it is noted that to see any noticeable change in the properties of an FRP composite as a function of temperature, the FRP must

be conditioned in an aqueous environment and not dry [24]. Because the baseline condition for the FRP composite conditioned in the aqueous environment is a fully-saturated coupon at room temperature, the predicted service life of the FRP is for an FRP that is used in a saturated or at least highly moist state. The service lives of FRPs used in dry conditions might be higher.

The specification provides for a range of 4 temperatures of conditioning, based on the glass transition temperature of the FRP composite material. The highest conditioning temperature is based on the premise that it is inappropriate to condition the material at a temperature too close to the FRP's wet glass transition temperature. At temperatures approaching the wet glass transition, it is likely that a change in mode of degradation will occur, thus invalidating one of the key assumptions of the Arrhenius model. Because the wet glass transition temperature is difficult to establish, the maximum conditioning temperature is taken to be 0.8 times the nominal glass transition temperature based on data provided by Chateauminois et al. [25]. The remaining 3 temperatures are then taken in even increments down to a temperature of 104 °F (40 °C).

Mechanical testing in tension and in short beam shear is completed after 28, 56, 112 and 224 days of conditioning in de-ionized water. The specification allows for flexural testing in the place of tension testing for Procedure B. Previous work has noted that gripping of conditioned specimens in tension fixtures is problematic [26]. Since what is of interest here is the rate of change in material properties as a function of time and temperature, it is deemed acceptable to use the easier-to-complete flexural tests as long as specimen geometry allows for such testing. It should be noted that the baseline flexural testing must be completed in Procedure A if it is to be used in Procedure B.

Procedure B requires that regression lines of property retention vs. log time must have an $R^2=0.80$ or greater in order to continue with the processing of the data via the Arrhenius model. If these regression relationships are not reasonably linear, then it does not make sense to make service life predictions based on the data. Such lack of linearity is not necessarily an indication that the material being tested is of poor quality. The only conclusion that can be made is that the pattern of degradation does not meet the Arrhenius assumptions. In any case, the property retention data at the prescribed times and temperatures should be provided as required in the reporting requirements. The requirement for an R^2 value of 0.80 or greater is the same as that given in ASTM D 3045, Standard Practice for Heat Aging of Plastics without Load [4].

At this time, it is difficult to qualify materials based on the results of these long-term predictions. Therefore, the specification requires that the information from the accelerated test regime must be provided but does not

attempt to judge whether the material 'passes' or 'fails' based on the results of these tests. It is anticipated that the requirement that this information be provided in a standard format, and using a standard procedure, will have the following benefits: (1) it will allow purchasers and manufacturers to identify constituent materials that have the best long-term properties; (2) it will allow purchasers to select the most durable materials from a given range of options; and (3) it will allow the authors of FRP design provisions to develop appropriate partial safety factors (ϕ factors) for material changes as a function of time in service. As more long-term data becomes available in the future, it is anticipated that limits will be set for Procedure B tests.

A note in the specification describes one potential application for the Procedure B data at this time. It may be possible, for certain materials, to specify a minimum property retention value, expressed as a percentage of the time zero (t_0) value, required for a desired service life. So, for example, a special provision for a given project might require that FRP materials provided as concrete reinforcements have a minimum tensile strength retention of 70% for a 50 year service life. The difficulty in requiring such provisions in the current specification is that the data are not available to determine the acceptable thresholds. In addition, it is not clear how such a provision could be enforced if the data from the accelerated testing regime was unable to be fit into the Arrhenius plots.

6.3.1. Optional environments for Procedure B

Material purchasers may be interested in material degradation in environments other than de-ionized water. To this end, two standard salt solutions are provided as optional environments. An environment that is representative of concrete pore water is also suggested as an optional environment. The specification writers suggest that the de-ionized water environment is the most appropriate for use at this time, and that other environments may be used when the accelerated testing methodology is further developed and verified. In many situations, conditioning in de-ionized water may be more damaging to the FRP than salt or alkaline solutions. d'Almeida has shown that moisture uptake in composites is quicker and reaches a higher equilibrium level when using de-ionized water [27]. He attributes this finding to the size of the ions in the salt solution, and the fact that these large ions impede moisture uptake in the composite.

7. Testing for part acceptance

Laminates are qualified and parts are accepted. Qualification testing has been defined and discussed previously. Acceptance testing can be thought of as proving that the parts being produced are made of the qualified

materials that the manufacturer developed for those parts. Because this specification covers a wide range of FRP composite forms, from flat sheets used for concrete repair to complex bridge deck sections, there are two different ways of thinking about the acceptance tests. For simple parts, there exists a one-to-one mapping between the qualification laminate and the production part. For example, a one-half inch diameter smooth rod might be used for qualification and a No. 4 FRP reinforcing bar might be produced as a product for acceptance testing. For complex parts, such as structural shapes, there might exist many different qualified laminates within a given part. This mapping is depicted in the example given in Annex B2 of the specification. In this instance, the part is accepted only after all of the laminates within the part are accepted. It is not permitted that a complex part be accepted by only testing one flange or one web within the part. Rather, all unique laminated constructions within the part must be sampled and tested according to Section A9 of the specification.

7.1. Procedure C—short-term material properties for acceptance

Acceptance tests under short-term ambient conditions (which may also be described as the ‘as-produced’ state) are included to ensure that the materials from the production process are ‘substantially’ the same materials as those that were qualified originally in Procedure A. If the coupon data from the reduced battery of acceptance tests is substantially the same as the data from the corresponding qualification tests, then designers can have confidence using the wide range of material property data established in the qualification testing (Procedure A) for design. Acceptance testing is limited to key mechanical and physico-chemical tests. All mechanical testing is in the longitudinal direction, so that coupons can be easily cut from production parts.

The allowable deviations, given in Table A.9.1 of the specification, are plus and minus the baseline values established in Procedure A. Strength values are allowed to deviate to a greater degree than modulus values, as it is generally known that strength changes (which depend on microscale variations in the material) are more likely than are modulus changes within a given material. The comparisons between qualification data and acceptance data are based on the testing of a minimum of 5 coupons for each test (qualification test and corresponding acceptance test). Without knowing the coefficient of variation (COV) of the given material/test method beforehand, it is impossible to say whether the variations presented in Table A.9.1 are statistically significant. The allowable 10 and 20% deviation ranges (i.e. ± 5 and $\pm 10\%$) were established based on review of test data from research on the accelerated testing of a range of FRP composite materials [24]. It may be

appropriate to tighten or loosen these ranges based on round-robin testing or review of additional test data that is undertaken specifically to validate these provisions.

7.2. Procedure D—short-term material properties for acceptance in the saturated state

Procedure D in the specification requires samples from the production part to be tested after substantial saturation in de-ionized water at 50 °C (122 °F). The changes in material properties after this conditioning are limited as outlined in Table A.9.2 of the specification. MIL Handbook 17 provides an excellent discussion of the rationale for screening of materials in a saturated condition [3]:

Most polymeric materials, whether unreinforced resin, polymeric composite matrix or a polymer-based fiber, are capable of absorbing relatively small but potentially significant amounts of moisture from the surrounding environment. The physical mechanism for moisture mass change, assuming there are no cracks or other wicking paths, is generally assumed to be mass diffusion following Fick's Law... Fickian moisture diffusion into or out of the interior occurs relatively slowly; many orders of magnitude slower than heat flow in thermal diffusion. Nevertheless, given enough exposure-time in a moist environment, a significant amount of moisture may be absorbed into the material. This absorbed moisture may cause material swelling, and, particularly at higher temperatures, may soften and weaken the matrix and matrix/fiber interface, which is deleterious to many mechanical properties that are often design drivers for structural applications.

The combination of moisture and elevated temperature has been found to be even more deleterious to composite material properties than either condition individually [3]. The ANSI specification for reinforced plastic ladders requires conditioning at 212 °F (100 °C) immersed in water [1], and the MIL-17 handbook specifies, among other types of elevated temperature/moisture conditions, a conditioning scheme of 160 °F (71 °C) immersed in water [28]. If materials are shown to retain most of their mechanical properties after short-term conditioning at elevated temperature and moisture, then they are likely to be durable in a wide range of service environments. The choice of 122 °F (50 °C) for conditioning in the FRP materials specification was made for two reasons. First, this temperature is safely below the conditioned glass transition temperature specified in Section A9.3.4 (160 °F/71.1 °C), ensuring that the material will not be subjected to a change of degradation mechanism. Second, 122 °F (50 °C) is seen as a reasonable upper bound to the service temperatures that can be expected for FRP materials in highway bridge applications.

Dramatic changes in mechanical properties after conditioning at elevated temperature and moisture are an indication of production and processing problems with the composite (Hawkins et al., 1998). Because Procedure D tests are completed on production materials, it is anticipated that this short-term conditioning and test-

ing will identify and reject composites that have not been properly processed. Wet, elevated temperature environments often trigger large moisture uptake and/or breakdown at the fiber/matrix interface in composites that have not been processed properly.

Finally, it is not yet possible to provide limits on the test data developed as part of Procedure B (the accelerated testing provision of the specification). Therefore, it is felt that some conditioned material screening function must take place at this time to ensure that only high quality composites are accepted as meeting the specification.

The limitations on property changes based on saturated conditioning are based on provisions in ICBO AC 125 and on property data presented by Gentry et al. and by Hawkins et al. [2,24,29]. Because type 2, 3 and 4 laminates have lower fiber volume fractions, and more potential for moisture absorption, it is anticipated that their properties may fall more as a result of conditioning to the saturated state. In addition, the use of multi-axis fabrics and continuous filament mats often lead to increased void content, which subsequently leads to reduced properties in the saturated state.

7.2.1. Sampling requirements and witness laminates for acceptance testing

For acceptance testing, the specification requires that all locations within a part that contain an instance of a specific qualified laminate shall be sampled and tested. For complex parts, it may be advisable to extract a sample set larger than that required to accept the laminate, and then to select coupons for testing from this set at random. Annex B2 of the specification provides a brief example of the sampling and testing required on a part that contains more than one distinct laminate.

In some cases, where the FRP composite is produced directly onto a substrate, it may be impossible to sample that laminate for acceptance testing. In such cases, the production of a witness laminate is acceptable. A witness laminate is typically produced in applications like on-site repair of concrete structures using FRP composites. In many cases, the additional specialized field tests, which are beyond the scope of this specification, are required to verify the bond between the FRP composite and the substrate.

8. Certification and reporting

The specification calls for the manufacturer to provide a document certifying that each lot of the material provided meets the specification. The specification does not dictate the manufacturer's quality control procedures. The certification is a formal acknowledgement by the manufacturer that such procedures are in place and that all lots of the material supplied are certified as meeting

the requirements of this specification. Purchasers who require testing on specific lots of material can invoke quality assurance elements of the specification at the time the material is ordered.

The manufacturer is required to present a summary of material constituent and processing information (see the discussion of materials and manufacturing earlier in this text). This information is needed so that traceability of the resulting FRP composite is possible. The test report also includes complete test results from representative parts taken from production lots of materials (Procedure C and Procedure D testing), and from the originally qualified laminates (Procedure A and Procedure B testing).

9. Quality assurance

The specification does not dictate the frequency with which sampling and testing according to Procedures C and D takes place. It is left to the manufacturer to determine the appropriate period for sampling and testing. Different production processes have different potential for error, and the manufacturer is expected to understand, monitor and control these processes. Development of quality control plans is typically completed as manufacturers apply for external quality certification such as ISO 9000. The manufacturer may elect to implement on-line process monitoring, non-destructive evaluation or additional quality control testing that is beyond the scope of this document.

In some instances, it may be advisable to supplement the materials screening provisions of the specification (Procedure A, Procedure D) with additional testing on samples taken from each production lot of material. This will ensure that the quality of the material remains consistent from lot-to-lot. Therefore, the specification provides an optional quality assurance provision, which allows purchasers to specify additional testing of each lot of material that is to be shipped for a given job. The quality assurance testing may be completed by the manufacturer, or coupons may be supplied to an independent laboratory for testing. If the manufacturer completes the testing, then notarized testing reports should be required at the time that the material is delivered. These reports are commonly provided as 'mill certs' with steel concrete reinforcing bars and with high strength structural bolts.

The quality assurance test reports differ from the previous test reports, as the reports described in the previous section document the full range of testing, from laminate qualification to part acceptance. The extent of sampling and number and types of tests used for quality assurance are left to the purchaser. As a minimum, it is suggested that the subset of mechanical and physical tests represented by Procedure C testing (Table A.9.1 of

the specification) be used by the purchaser for quality assurance.

10. Summary and conclusions

This article highlights key features of a model specification for fiber reinforced polymer (FRP) materials for civil engineering. The specification has been developed to provide a common framework for the testing of FRP composites and for the reporting of test data. Classification of FRP laminates is required so that similar laminated constructions can be tested in an identical manner. Reporting of constituent materials is required, and the admissibility of certain key constituents (fibers, resins and additives) is limited where the use of these constituents may affect quality and durability. To aid designers and end users who seek to develop design values for the materials with appropriate levels of safety, the specification requires the reporting of an extensive set of material characterization data in the 'as-produced' and in the 'hot-wet' conditioned state. The specification requires the completion of an accelerated conditioning and testing regime, with provisions for prediction of retained strengths where data warrant such a prediction. These retained strength predictions, given as a function of service life and service temperature, can subsequently be used to determine partial safety factors for environmental exposure. Finally, the specification contains quality assurance provisions that allow the end user to require testing on individual lots of material provided for a given project.

Like all standards and specifications, the proposed model specification is intended to be a continually evolving document and will reflect the state-of-the-art at a given stage of development. The authors welcome the discussion on the proposed model specification from all interested parties. Input from end-users, government officials and industry professionals is being continuously solicited to improve and refine the model specification.

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Appendix A: Model specification for FRP composites

A.1. Scope

A.1.1. General. This specification provides procedures for classifying, testing, qualifying and accepting fiber reinforced polymer (FRP) composite materials for use in highway bridges. The specification covers FRP composite materials used for a wide variety of highway structural applications, including structural shapes; rods, bars and grids used for internal concrete reinforcement; strips, plates and shells used for external reinforcement for concrete, wood, steel or masonry; and stay-in-place formwork.

A.1.2. Constituents and classification. FRP composite materials using isophthalic polyester, vinylester and epoxy resins, and glass or carbon fibers are covered by the specification.¹ FRP composite materials are classified based on resin type, fiber type and fiber architecture. Any production method for FRP composite laminates and FRP parts is permitted as long as minimum property requirements are satisfied.

*A.1.3. Qualification testing*². FRP composite materials are qualified based on testing and assessment of coupons cut from FRP parts or from rectangular panels or compact shapes produced specifically for qualification. Procedures are provided for determining a set of mechanical and physical properties for subsequent development of design values.³ Key mechanical and physical properties must meet minimum requirements (Procedure A). A procedure for predicting the long-term properties of the qualified FRP composite material based on the acceleration of degradation at elevated temperatures while immersed in water is also stipulated (Procedure B). A service life prediction procedure based on the Arrhenius relationship is provided.

A.1.4. Acceptance testing. FRP composite parts produced according to this specification must be constructed of one or more qualified FRP composite laminates. Each distinct laminate within the FRP composite part must be sampled and tested to accept the part. Coupons cut from actual parts are subject to a reduced set of

¹ The specification has been organized in a manner that will allow for the inclusion of additional fiber and resin types as long-term performance data for these materials become available.

² See Annex B1 for a schematic depicting the qualification and acceptance procedures presented in this specification.

³ Design codes such as the AASHTO Standard Specification for Highway Bridges use strength data from material testing to arrive at design allowable values for materials. Existing design guides for FRP composites such as the American Concrete Institute document ACI 440.1R-01 'Guide to the Design and Construction of Concrete Reinforced with FRP Bars' use mechanical property data for FRPs to determine design allowable stresses.

mechanical and physical tests in the as-received condition (Procedure C). Data from these tests must be in substantial agreement with data obtained in the qualification tests. Acceptance tests are also conducted after conditioning the laminates in de-ionized water at 50 °C (122 °F) until they are substantially saturated (Procedure D). These tests are included to ensure that the laminates do not lose substantial strength in hot, wet conditions, as these conditions may be anticipated to occur in the highway environment.

A.1.5. Sampling and testing. All material properties are determined from ASTM standard test methods on coupons cut from FRP composite laminates or parts. In some cases (e.g. small rods or bars), the full section may be tested if its dimensions are appropriate for the ASTM test being performed. Full-section testing of complex FRP composite parts is not covered by this specification.⁴

A.1.6. Reporting and certification. Results from the test methods stipulated in this specification shall be reported in a standardized manner. Materials represented as meeting the requirements of this specification must be certified as conforming to the requirements given herein.

A.2. Referenced documents

A.2.1. ASTM standards.

B 117	Standard practice for operating salt spray (Fog) apparatus
C 162	Standard terminology of glass and glass products
C 219	Standard terminology relating to hydraulic cement
C 666	Standard test method for resistance of concrete to rapid freezing and thawing
C 859	Standard terminology relating to nuclear materials
C 904	Standard terminology relating to chemical-resistant non-metallic materials
D 123	Standard terminology relating to textiles
D 256	Standard test methods for determining the izod pendulum impact resistance of plastics
D 570	Standard test method for water absorption of plastics
D 618	Standard practice for conditioning plastics for testing
D 638	Standard test method for tensile properties of plastics
D 648	Standard test method for deflection temperature of plastics under flexural load in the edgewise position
D 695	Standard test method for compressive properties of rigid plastics
D 696	Standard test method for coefficient of linear thermal expansion of plastics between (30 °C and 30 °C with a vitreous silica dilatometer
D 790	Standard test methods for flexural properties of unreinforced and reinforced plastics and electrical insulating materials

D 792	Standard test methods for density and specific gravity (relative density) of plastics by displacement
D 883	Standard terminology relating to plastics
D 907	Standard terminology of adhesives
D 1141	Standard practice for substitute ocean water
D 1929	Standard test method for determining ignition temperature of plastics
D 2344	Standard test method for short-beam strength of polymer matrix composite materials and their laminates
D 2583	Standard test method for indentation hardness of rigid plastics by means of a Barcol impressor
D 2584	Standard test method for ignition loss of cured reinforced resins
D 2990	Standard test methods for tensile, compressive, and flexural creep and creep-rupture of plastics
D 3039	Standard test method for tensile properties of polymer matrix composite materials
D 3045	Standard practice for heat aging of plastics without load
D 3171	Standard test method for constituent content of composite materials
D 3410	Standard test method for compressive properties of polymer matrix composite materials with unsupported gage section by shear loading
D 3479	Standard test method for tension–tension fatigue of polymer matrix composite materials
D 3846	Standard test method for in-plane shear strength of reinforced plastics
D 3878	Standard terminology composite materials
D 3916	Standard test method for tensile properties of pultruded glass-fiber-reinforced plastic rod
D 3918	Standard terminology relating to reinforced plastic pultruded products
D 4175	Standard terminology relating to petroleum, petroleum products and lubricants
D 4475	Standard test method for apparent horizontal shear strength of pultruded reinforced plastic rods by the short-beam method
D 4476	Standard test method for flexural properties of fiber reinforced pultruded plastic rods
D 4502	Standard test method for heat and moisture resistance of wood-adhesive joints
D 5083	Standard test method for tensile properties of reinforced thermosetting plastics using straight-sided specimens
D 5229	Standard test method for moisture absorption properties and equilibrium conditioning of polymer matrix composite materials
D 5379	Standard test method for shear properties of composite materials by the v-notched beam method
E 6	Standard terminology relating to methods of mechanical testing
E 122	Standard practice for calculating sample size to estimate, with a specified tolerable error, the average for a characteristic of a lot or process
E 631	Standard terminology of building constructions
E 632	Standard practice for developing accelerated tests to aid prediction of the service life of building components and materials
E 831	Standard test method for linear thermal expansion of solid materials by thermomechanical analysis
E 1356	Standard test method for assignment of the glass transition temperatures by differential scanning calorimetry or differential thermal analysis
E 1640	Standard test method for assignment of the glass transition temperature by dynamic mechanical analysis
E 2092	Standard test method for distortion temperature in three-point bending by thermal mechanical analysis

⁴ The overall performance of an FRP composite assembly or part may not be completely predicted by testing coupons taken from the part. Elastic stability, fatigue behavior, and strength failures at part junctions may not be uncovered by testing coupons only. Therefore, full-section testing of FRP parts may be required in addition to the coupon-level testing outlined in this document.

A.2.2. *Other standards*: ‘American National Standard for Ladders – Portable Reinforced Plastic.’ American National Standards Institute (ANSI) A14.5-2000.

A.3. Terminology

A.3.1. *Definitions*. *Acceptance test* – a test, or series of tests conducted by the procuring agency, or an agent thereof, upon receipt to determine whether an individual lot of materials conforms to the purchase order or contract or to determine the degree of uniformity of the material supplied by the vendor, or both (ASTM D 907).

Additive – a material added to another, usually in small amounts to impart or enhance desirable properties or to suppress undesirable properties (ASTM D 4175).

Balanced laminate – any laminate that contains one ply of minus theta orientation with respect to the laminate principal axis for every identical ply with a plus theta orientation (ASTM D 3878).

Bar – a round, square, rectangular or other polygonal solid member having a length greater than its width or thickness (ASTM E 631).

Binder – in reinforced plastic, the continuous phase which holds together the reinforcement (ASTM D 883). See also matrix.

Commercial off-the-shelf (COTS) parts – parts whose laminates have been pre-qualified, which have successfully completed acceptance testing, and are generally stocked by the composites manufacturer.

Composite material – a substance consisting of two or more materials, insoluble in one another, which are combined to form a useful engineering material possessing certain properties not possessed by the constituents (ASTM D 3878); *n* – a solid product consisting of two or more distinct phases, including a binding material (matrix) and a particulate or fibrous material (ASTM D 883).

Continuous filament – an individual rod of glass (or carbon) of small diameter, which is flexible and of great or indefinite length.

Creep – the time-dependent increase in strain in a solid resulting from force. Creep tests are usually made at constant load and at constant temperature. For tests on plastics, the initial strain however defined, is included and for metals, it is not. This change in strain is sometimes referred to as creep strain (ASTM E 6).

Creep rupture stress – the stress that will cause fracture in a creep test at a given time, in a specified constant environment (ASTM E 6). This is sometimes referred to as the ‘stress-rupture strength.’ In glass technology this is referred to as the ‘static fatigue strength.’

Crosslinking – the formation of a three-dimensional polymer by means of interchain reactions resulting in changes in physical properties (ASTM D 883).

Cure – to change the properties of a polymeric system into a more stable, usable condition by the use of heat, radiation or reaction with chemical additives (ASTM D 883).

Epoxy resin – a viscous liquid or brittle solid containing epoxide groups that can be crosslinked into final form by means of a chemical reaction with a variety of setting agents used with or without heat (ASTM C 904).

Fatigue life – the number of cycles of stress or strain of a specified character that a given specimen sustains before failure of a specified nature occurs (ASTM E 6).

Fiber architecture – arrangement of fibers within the FRP composite material.

Fiber content – the amount of fiber present in a composite expressed either as percent by weight or percent by volume. This is sometimes stated as a fraction, that is, fiber volume fraction (ASTM D 3878).

Fiber volume fraction – see fiber content.

Filament – see continuous filament.

Filler – in composite materials, a primarily inert solid constituent added to the matrix to modify the composite properties or to lower cost (ASTM D 3878); *n* – a relatively inert material added to a plastic to modify its strength, permanence, working properties or other qualities, or to lower costs (ASTM D 883).

Flash ignition temperature (FIT) – the minimum temperature at which, under specified test conditions, sufficient flammable gases are emitted to ignite momentarily upon application of a small external pilot flame (ASTM D 1929).

Freeze thaw – alternately lowering the temperature of specimens from 40 to 0 °F (4.4 to –17.8 °C) and raising it from 0 to 40 °F (–17.8 to 4.4 °C) in not less than 2 h nor more than 5 h (ASTM C 666).

Gel coat – a gelled resin layer added to the mold prior to lay-up on the surface, which then becomes an integral part of the composite material. Used to provide a smooth surface and protect the fiber reinforcements.

Glass transition – the reversible change in an amorphous polymer, or in amorphous regions of a partially crystalline polymer, from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one (ASTM D 883).

Glass transition temperature (T_g) – the approximate midpoint of the temperature range over which the glass transition takes place (ASTM D 883).

Hybrid laminate – a laminate using more than one filament type (carbon or glass) in its plies. As used in this specification, up to 20% by volume of the total fiber content of a hybrid laminate may be of the secondary fiber type.

Hydraulic cement – a cement that sets and hardens by chemical interaction with water and that is capable of doing so under water (ASTM C 219).

Ignition temperature – the temperature at which a material or its pyrolysis products can be ignited under

given conditions of temperature, pressure and oxygen concentration (ASTM D 1929).

Isophthalic polyester resin – a thermosetting resin that results from the products of an isophthalic acid – glycol reaction blended with a monomer.

Laminate – any fiber- or fabric-reinforced composite consisting of laminae (plies) with one or more orientations with respect to some reference direction (ASTM D 3878).

Lay up – in reinforced plastics, to assemble layers of resin-impregnated material for processing (ASTM D 883).

Longitudinal axis – axis selected for coupon sampling and testing. Must be either the primary structural direction of the part or the primary direction of production for the part.

Longitudinal direction – along the longitudinal axis.

Mat – a fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder (ASTM D 883).

Matrix – the essentially homogenous phase in a composite material in which reinforcements such as fibers, filaments, particles etc., are embedded (ASTM E 631).

Mechanical properties – those properties of a material that are associated with elastic and inelastic reaction when force is applied, or that involve the relationship between stress and strain (ASTM C 859).

Moisture saturation content – the moisture equilibrium content at the maximum possible moisture exposure level, wherein the material contains the greatest possible amount of absorbed moisture (ASTM D 5229).

Part – a single-piece FRP composite element, for which all the resin in the FRP composite is placed and cures at the same time.

Physical properties – material properties that involve neither chemical change nor the stress- and strain-related properties described as mechanical properties.

Plate – a flat, rolled sheet having a width and length much greater than its thickness (ASTM E 631).

Polymer – a substance consisting of molecules characterized by the repetition (neglecting ends, branch junctions and other minor irregularities) of one or more types of monomeric units (ASTM D 883).

Pre-accepted parts – parts whose laminates have previously passed qualification tests that have also passed acceptance tests as described in this specification.

Pre-qualified laminates – laminates that have undergone and passed qualification tests as described in this specification.

Processing method – process by which reinforcing fibers, resin and any fillers or additives are combined together and cured into a composite material.

Production lot – that part of one manufacturer's production made from the same nominal raw material

under essentially the same conditions and designed to meet the same specifications (ASTM D 123).

Pultrude – to draw resin-impregnated reinforcement through a die (ASTM D 3918).

Qualification test – a series of tests conducted by the procuring agency, or an agent thereof, to determine conformance of materials, or materials system, to the requirements of a specification which normally results in a qualified products list under the specification (ASTM D 907).

Reinforcement – a strong inert material bonded into a plastic to improve its strength, stiffness and impact resistance. Reinforcements are usually long fibers of glass, asbestos, sisal, cotton, etc., in woven or non-woven form.

Resin – a solid, semisolid or pseudo-solid organic material that has an indefinite and often high molecular weight, exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and usually fractures conchoidally (ASTM D 3878).

Rod – see bar.

Roving – in fibrous composites, a large filament count tow (ASTM D 3878); *n* – in glass textiles, a multiplicity of filaments or yarns gathered together into an approximately parallel arrangement without twist (ASTM C 162).

Strand – a multiplicity of continuous glass filaments combined into a single compact unit, without twist (ASTM C 162).

Symmetric laminate – a laminate in which the stacking sequence for the plies located on one side of the geometric midplane are the mirror image of the stacking sequence on the other side of the midplane (ASTM D 3878).

Test property laminate – a rectangular laminate manufactured for mechanical testing using the same resin, laminate structure and production method as the part in which it is to be used.

Thermoset – a class of polymers that, when cured using heat, chemical or other means, changes into a substantially infusible and insoluble material (ASTM D 3878); *n* – a plastic that, after having been cured by heat or other means, is substantially infusible and insoluble (ASTM D 883).

Thermosetting resin – a polymeric material capable of crosslinking under the influence of heat, pressure, radiation, ultraviolet light or chemical agents to form a thermoset (ASTM D 907).

Tow – in fibrous composites, a continuous, ordered assembly of essentially parallel, collimated filaments, normally without twist and of continuous filaments (ASTM D 3878).

Transverse axis – structural axis at 90° to the longitudinal axis and in the plane of the laminate.

Unidirectional fiber-reinforced composite – any fiber-reinforced composite with all fibers aligned in a single direction (ASTM D 3878).

Veil – an organic or inorganic fiber mat added to the surface of a part to produce a smooth surface.

Vinyl ester resin – a thermosetting reaction product of epoxy resin with a polymerizable unsaturated acid, usually methacrylic acid, which is then diluted with a reactive monomer, usually styrene (ASTM C 904).

Volume fraction – see fiber content.

Witness laminate – a rectangular laminate produced with the same resin, fillers and resin initiation system, the same reinforcement fibers and fabrics, on the same substrate, and using the same production method, and the same cure conditions as the original part that it witnesses for. Also sometimes referred to as a witness panel.

0° direction – see longitudinal axis.

A.4. Classification

A.4.1. Composite materials covered by this specification are classified according to resin type, fiber type and fiber architecture.⁵ The classification system described below applies to each distinct laminate within a given part.⁶ For parts of small cross-section such as rods, the entire part may be classified as one laminate type. For molded parts or continuously produced parts with complex cross-sections, the part may be comprised of multiple laminated elements, each of a different classification. Each of these distinct laminates within the overall part shall be classified according to the procedure outlined in this section and shall undergo a complete set of qualification and acceptance tests as described in Section A.8 and A.9 of this specification.

A.4.2. *Resin type*. Any commercial grade of isophthalic polyester, vinylester or epoxy thermosetting resin, subject to the limitations given in Section A.6.2, is permitted. FRP composites using isophthalic polyester resin shall be identified as type *P*; those using vinylester resin shall be identified as type *V*; those using epoxies shall be identified as type *E*.

A.4.3. *Fiber type*. Any commercial grade glass fiber⁷ (designated *G*) or carbon fiber (designated *C*) is

⁵ Fiber architecture describes the way that the individual reinforcing fibers are collected into fiber bundles or woven into fabrics, and subsequently, how these bundles or fabrics are placed into the FRP part. A laminate is a plate-like structure with a unique thickness that is significantly smaller than the other two dimensions. A given FRP part might contain many different laminates. Rods or compact shapes such as squares often have only unidirectional reinforcement and do not contain distinct laminae. Such constructions are covered by the specification and are classified as Type 1 laminates.

⁶ As defined and used in this specification, an FRP part is produced in a single operation and is not a collection of sub-components that have been bonded together after the polymer has cured.

⁷ Glass fibers shall be of E-glass, S-glass or derivatives of these types. This restriction is not intended to prevent the use of newer glass formulations intended to prevent corrosion of structural glass fibers.

permitted. The fiber may be in the form of unidirectional rovings or tows of any size or weight, or can be in the form of stitched, woven, braided or non-woven fabrics, or mats of any size or weight.

A.4.4. *Hybrid composites*. Mixing of fiber types (glass and carbon) within an individual laminate is permitted as follows. Up to 20% of the total fiber volume in a carbon-fiber laminate may be glass fiber. Likewise, up to 20% of the total fiber volume in a glass-fiber laminate may be carbon fiber. No restrictions are placed on the orientation of the secondary fiber within the hybrid laminate. However, the classification categorization described in Section 4.5 still applies.⁸ The hybrid laminate is classified as type *CH* if it is a carbon fiber laminate with up to 20% glass fiber or a type *GH* if it is a glass-fiber laminate with up to 20% carbon fiber.

A.4.5. *Fiber architecture and laminate type*. Four types of laminated constructions are defined according to the direction of the continuous reinforcing fibers relative to the longitudinal axis of the laminate.⁹ The longitudinal axis of the laminate shall be as defined in Section A.9.4.2.1.

A.4.5.1. *Type 1 laminate*—A Type 1 laminate shall have a total fiber volume fraction of 50% or greater.¹⁰ For Type 1 laminates, 95% or more of this fiber shall be continuous fiber in the direction of the longitudinal axis.

A.4.5.2. *Type 2 laminate*—A Type 2 laminate shall have a total fiber volume fraction of 40% or greater. For Type 2 laminates, 75% or more of this fiber shall be continuous fiber in the direction of the longitudinal axis.

A.4.5.3. *Type 3 laminate*—A Type 3 laminate shall have a total fiber volume fraction of 40% or greater. For Type 3 laminates, 40% or more of this fiber shall be continuous fiber in the direction of the longitudinal axis.¹¹

⁸ Hybrid laminates are permitted only to the extent that the hybrid laminate is a dominantly carbon fiber or dominantly glass-fiber laminate. Minimum mechanical properties for *G* or *C* classifications, given in Table A.8.2, still apply to the hybrid laminate.

⁹ It is recognized that many production methods do not produce a layered structure typical of a part that is molded with many layers of woven or non-woven fabrics. The internal structure is still referred to as a laminate even if no distinct layering exists.

¹⁰ Unidirectional rods, FRP concrete reinforcements and pre-stressing tendons are typically Class 1 laminates.

¹¹ A laminate having no continuous fiber in the longitudinal direction does not qualify as a Type 3 laminate. Therefore, for example, a purely filament wound tube with no longitudinal fiber or a purely $\pm 45^\circ$ laminate is not an acceptable laminate according to this specification. In the large majority of civil engineering structural elements, applied loads are carried by normal stresses along major axes of the part. Therefore, the Type 3 laminate requires a minimum volume fraction of 40% continuous fiber in the direction of the longitudinal axis.

A.4.5.4. Type 4 laminate—A Type 4 laminate shall have a total fiber volume fraction of between 30 and 50%. For Type 4 laminates, 95% or more of this fiber shall be continuous fiber in the direction of the longitudinal axis.¹²

A.4.5.5. Fibers considered as being in the direction of the longitudinal axis are continuous fibers inserted into the structural part as rovings, tows, woven fabrics or stitched fabrics. Chopped strand or continuous filament mats are not considered to contribute to the longitudinal axis fiber. However, these mats are considered to contribute to the total fiber volume fraction.

A.4.5.6. Laminate type, according to Sections A.4.5.1–A.4.5.4, shall be reported by the material manufacturer and need not be established by testing. However, the fiber volume fraction must be verified according to the testing requirements for volume fraction given in Section A.8.

A.4.6. *Classification.* The FRP composite material is classified on the laminate level according to its fiber type, resin type and fiber architecture. The FRP composite laminate shall have a layup that does not cause in-plane axial-shear or out-of-plane axial-bending coupling. This may be achieved with the use of unidirectional laminates or balanced and symmetric laminates.¹³ The classification system is applied to every distinct laminate thickness and fiber architecture within the FRP part. The classification nomenclature is as follows: fiber type, polymer resin type, fiber architecture/laminate type (e.g. GV2 designates a glass/vinylester Type 2 FRP composite laminate). Sections A.8 and A.9 of this specification contain qualification and acceptance criteria for the following classes of composite laminates.¹⁴

A.4.6.1. GP1, GV1 and GE1;

A.4.6.2. GP2, GV2 and GE2;

A.4.6.3. GP3, GV3 and GE3;

A.4.6.4. GP4, GV4 and GE4;

A.4.6.5. CP1, CV1 and CE1;

A.4.6.6. CP4, CV4 and CE4;

A.4.6.7. Hybrid laminates with major fiber type as listed above, for example GHE1 and CHE4.

¹² The Type 4 laminate classification is reserved primarily for low volume fraction laminates such as those that are laid-up in the field using manual laminate consolidation techniques. Most machine-produced laminates will have higher volume fractions and higher mechanical properties and will be more properly classified as Type 1, Type 2 or Type 3 laminates.

¹³ A balanced laminate has no axial-bending coupling. A symmetric laminate has no axial-shear coupling.

¹⁴ Composite laminates not explicitly listed in this section are not widely produced for application in highway structures. Therefore, little data currently exists to determine qualification or acceptance limits for these laminates and they are not included in the specification at this time. In particular, note that types CP2, CV2, CE2, CP3, CV3 and CE3 are not included at this time, due to a lack of data regarding the material properties of these laminate types.

A.5. Ordering information

A.5.1. *Scope.* This section describes procedures for ordering FRP composite materials produced to meet this specification.

A.5.2. *Ordering of pre-accepted parts.* Pre-accepted parts are parts that are constructed of pre-qualified laminates that have been tested according to Section A.8 of this specification. Acceptance testing of the parts per Section A.9 has also been completed. Testing reports for qualification and acceptance tests must be ‘on file’ for these parts. The parts may, therefore, be considered as ‘commercial off-the-shelf’ or COTS parts and may be delivered from stock as meeting this specification. No additional testing is required at the time of order placement. The manufacturer shall certify, in accordance with Section A.11.1, that the pre-accepted parts meet the requirements of this specification.

A.5.3. *Ordering of non pre-accepted parts constructed of pre-qualified FRP composite laminates.* Non pre-accepted FRP composite parts may be ordered to be constructed from pre-qualified laminates. In this instance, testing described in Section A.8 of this specification is not required. Laminate qualification testing data must be ‘on file’ and shall be delivered with the certification of the FRP composite material. Full testing of laminates cut from the actual parts, as described in Section A.9, is required.

A.5.4. *Ordering of non pre-accepted parts constructed of non pre-qualified laminates.* If FRP composite parts are to be constructed of non pre-qualified laminates, then laminate testing on qualification coupons, as described in Section A.8, and testing on laminates cut from production parts, as described in Section A.9, must be completed.¹⁵

A.5.5. Purchasers may specify optional independent quality assurance testing, as described in Section A.13, if desired. The extent of and responsibility for such testing shall be established at the time of ordering of the materials.

A.6. Materials

A.6.1. *Scope.* This section describes the constituent materials that are permitted in FRP composite parts. Limits on additives and fillers, relative to base resin content, are given in this section.

A.6.2. *Polymer material.* The base polymer must be 100% of the specified resin type (either isophthalic

¹⁵ Lead times for FRP composite parts that are produced from non pre-qualified laminates will take up to 1 year for delivery due to the length of time required to complete Procedure B testing.

polyester, vinylester or epoxy). Blending of resins of the same type is permitted. Blending of resins of different types is not permitted.

A.6.2.1. Isophthalic polyester resin is not permitted for use in FRP parts that will be in direct contact with, embedded in, or adhered to hydraulic cement concrete.¹⁶

A.6.3. Reinforcements: A.6.3.1. *Fiber material type*—The fiber material shall be a commercial-grade glass or carbon fiber.

A.6.3.2. *Fiber treatments*—Fiber treatments (sizings) shall be appropriate for the resin system used.

A.6.3.3. *Fiber form*—The following types of fiber bundles are allowed within the part:

A.6.3.3.1. One-dimensional (rovings or tows);

A.6.3.3.2. Two-dimensional (woven or non-woven);

A.6.3.3.3. Continuous fiber mats (continuous strand mats, continuous filament mats (CFM));

A.6.3.3.4. Non-continuous fiber mats (chopped strand mats (CSM)).

A.6.4. Fillers. Commercial grade inorganic fillers such as kaolin clay, calcium carbonate and alumina trihydrate may be used, but shall not exceed 20% by weight of the specified base polymer resin.

A.6.5. Additives. Additives making up part of the cure system or added for other reasons shall be limited as given below. All additives shall be appropriate for the fiber and resin system used.

A.6.5.1. *Coupling agents*—Coupling agents may be added as necessary. The amount is not limited.

A.6.5.2. *Release agents*—Release agents may be added as necessary. The amount is not limited.

A.6.5.3. *Initiators*—Initiators may be added as necessary. The amount is not limited.

A.6.5.4. *Hardeners*—Hardening agents may be added as necessary. The amount is not limited.

A.6.5.5. *Dilutents*—Dilutents such as styrene added to the base polymer at the time of production shall not exceed 10% by weight of the specified base polymer resin.

A.6.5.6. *Promoters*—Promoters may be added as necessary. The amount is not limited.

A.6.5.7. *Catalysts*—Catalysts may be added as necessary. The amount is not limited.

A.6.5.8. *UV agents*—UV agents may be added as necessary, or as agreed upon by the purchaser and manufacturer. The amount is not limited.

A.6.5.9. *Fire retardants*—Fire retardants may be added as necessary, or as agreed upon by the purchaser and manufacturer. The amount is not limited.

A.6.5.10. *Low-profile (shrink) additives*—Low-profile shrink additives shall not exceed 20% by weight of the specified base polymer resin.

A.6.5.11. *Foaming agents*—Foaming agents may be added as necessary. The amount is not limited.

A.6.5.12. *Pigments*—Pigments may be added as necessary. The amount is not limited.

A.6.6. Other. A.6.6.1. *Gel coats*—Gel coats may be used to produce a smooth surface on a composite part. Where gel coats are used, their thickness must be included in the measurement of laminate thickness. The thickness of the gel coat is not limited.¹⁷

A.6.6.2. *Veil*—Veils may be included in a laminate in a composite part to produce a smooth surface. Where veils are used, the thickness of the veil layers must be included in the measurement of laminate thickness. The thickness of the veil layer is not limited.

A.7. Manufacturing

A.7.1. The method of manufacturing includes the method of placing and arranging the fibers and fiber forms, the method of introducing the thermosetting resin into and around the fiber, and the method for activating the resin to produce a cured laminate.

A.7.2. Method of manufacture shall be any open or closed molded process that produces a fully cured composite.¹⁸ Either pressure molded¹⁹ or manual molded²⁰ processes are permitted.

A.7.3. *Production lot size*—Manufacturer shall provide a definition for lot size given the production method used for the FRP composite part. Manufacturer shall define an average value and range for the amount of material (by weight, square area, linear measurement, etc.) that is produced in one lot.

A.7.4. *Production changes*—If a change in constituent material (e.g. resin, fiber type, fabric weave, stacking sequence) or manufacturing process (e.g. pultrusion, vacuum molding, hand lay-up) is made on an FRP

¹⁷ The thickness of the gel coat and veil layers are not limited, however, the overall thickness of the laminate must include veil and gel coat layers. Therefore, the use of thick gel coats or veil layers will impact volume fractions.

¹⁸ The degree of cure of a polymer is difficult to measure. The intent of this specification is that FRP composites produced to meet this specification shall have a minimum of un-crosslinked monomer left in the part after production. The specification does not insist that the manufacturer demonstrate that the part is fully-cured, but does require that laminates within the part demonstrate a minimum glass transition temperature (T_g) and minimum Barcol hardness value.

¹⁹ Pressure-molded processes include pultrusion, resin-transfer molding and matched die.

²⁰ Manual molding processes include vacuum-assisted resin transfer molding (VARTM), Seeman's composite resin injection molding process (SCRIMP), hand lay-up.

¹⁶ Hydraulic cement concretes include Portland cement concretes and other concretes using an inorganic binder as defined by ASTM C 219.

composite part, then new laminates must be qualified for that part.

A.8. Physical and mechanical testing for laminate qualification

A.8.1. Scope. The physical and mechanical properties of each laminate that a manufacturer intends to use in one or more parts must be determined according to the provisions of this section. The manufacturer shall cut coupons from a standard part if said part provides sufficient longitudinal and transverse dimensions for complete laminate characterization. The laminate qualification testing may also take place on coupons cut from a rectangular plate described as a ‘Test Property Laminate.’²¹ The manufacturer shall provide physical and mechanical data for the laminate as described in Table A.8.1. The laminate must meet the minimum property requirements described in Procedure A, given in Section A.8.5. Finally, the long-term property retention data determined by Procedure B (Section A.8.6) must also be provided for the laminate. Any laminate that undergoes Procedure A and Procedure B testing, and meets the minimum property requirements given in Section 8.2 is denoted as a ‘qualified laminate’ and may be used in any part. The part itself must also be accepted according to Section A.9.

A.8.2. Laminate identification. The manufacturer shall provide a distinct identification code for each qualified laminate in the following form: CCC-TT.T-AAA, where CCC is the classification sequence (for example, VG2), TT.T is the laminate nominal thickness in millimeters (for example, 12.6), and AAA is an alphanumeric identifier (for example, A45).²²

A.8.3. Test property laminate. A test property laminate (TPL) may be produced for laminate qualification. The TPL is a rectangular panel using the same resin, laminate structure and production method that will be used for the part. If the production part, is a round rod or square bar, then the TPL may be round or square and have the same dimensions as the production part.²³ The TPL must be manufactured with the same materials and manufacturing processes as the laminate in the part for which it substitutes. For Type 2 and 3 laminates, the

TPL must be wide enough to allow for full longitudinal and transverse characterization per Table A.8.1.

A.8.4. Summary of testing requirements. The physical and mechanical properties of the laminate shall be determined according to two procedures: (1) Procedure A—Short-term material properties for qualification and (2) Procedure B—Long-term material properties for qualification. Procedure A requires that specific tests be conducted and that the results of these tests be reported in order for the material to meet this specification (see Table A.8.1). In addition, Procedure A provides material acceptance limits for FRP composite materials based on their classification (see Table A.8.2). Procedure B provides a method for predicting the long-term property retention of the materials using an accelerated aging procedure.

A.8.5. Procedure A—short-term material properties for qualification. **A.8.5.1. Scope**—Table A.8.1 lists the required tests, and denotes the allowable test methods.²⁴ Sampling procedures and the number of test specimens required for the various tests shall be in accordance with Section A.9.4.

A.8.5.2. Significance and use—Reporting of properties is required so that end users of the material will have data on the properties of the material for use in developing design values. The comparison of material properties with the minimums given in Table A.8.2 is intended to ensure that the laminates produced have the mechanical and physical properties that their laminate classification predicts that they should have. They are not intended as design values. Failure to meet these limits indicates that the laminate may not have appropriate quality for use as load-bearing members in highway bridge applications.²⁵

A.8.5.3. Sampling requirements—All test specimens shall be taken either from a standard part or from a TPL. The number and type of samples shall be in accordance with the ASTM test method for the specific property to be determined (see Table A.8.1). However, if the number of specimens is not specified in the test

²¹ In many cases, it is difficult to cut transverse direction coupons from actual parts (due to small dimensions in these directions). Therefore, the use of a test property laminate, which must be produced of sufficient dimensions so that longitudinal and transverse coupons can be cut, is permitted for laminate qualification.

²² The manufacturer’s identifier may use more than 3 letters and numbers if necessary.

²³ The use of round or square test property laminates (TPL) is limited to Class 1 or Class 4 FRP laminates for which there are no transverse property testing requirements.

²⁴ The specification identifies multiple tests for characterization of some properties. Where more than one test method is indicated, the testing may be completed using any of the ASTM tests indicated. All similar tests (e.g., tension, compression) must be completed using the same test method and the test method used must be reported along with the test data. In addition, the same test method must be used for both the qualification tests (Procedures A and B) and the acceptance tests (Procedures C and D). For tests on rods, only one of the listed test methods is appropriate.

²⁵ Minimum property values for FRPs in the various classes were arrived at after a survey of (1) reports in the archival literature on the testing of FRP materials, (2) manufacturer’s published mechanical test data, (3) Mil Handbook 17, Volume 2, and (4) mechanics calculations on trial laminates.

Table A.8.1

Short-term material properties for qualification (Procedure A)

	Material property	ASTM test method(s) Required ^{a,b}	Testing required by laminate type
Mechanical properties			
Strength properties	Longitudinal tensile strength	D 3039, D 5083, D 638, D 3916	All types
	Longitudinal compressive strength	D 3410, D 695	All types
	Longitudinal short beam shear strength	D 2344, D 4475	All types
	In-plane shear strength	D 5379, D3846	All types
	Impact resistance	D 256	All types
	Transverse tensile strength	D 3039, D 5083, D 638, D 3916	Type 2 and 3 only
	Transverse compressive strength	D 3410, D 695	Type 2 and 3 only
Stiffness properties	Transverse short beam shear strength	D 2344, D 4475	Type 2 and 3 only
	Longitudinal tensile modulus	D 3039, D 5083, D 638, D 3916	All types
	Longitudinal compressive modulus	D 3410, D 695	All types
	Major (longitudinal) poisson ratio	D 3039, D 5083, D 638	All types
	In-plane shear modulus	D 5379	Type 2 and 3 only
	Transverse tensile modulus	D 3039, D 5083, D 638, D 3916	Type 2 and 3 only
	Transverse compressive modulus	D 3410, D 695	Type 2 and 3 only
Physical properties			
	Fiber volume fraction ^c	D 3171, D 2584	All types
	Density	D 792	All types
	Barcol hardness	D 2583	All types
	Glass transition temperature ^d	E 1356, E 1640, D 648, E 2092	All types
	Water absorbed when substantially saturated ^e	D 570	All types
	Longitudinal coefficient of thermal expansion	E 831, D 696	All types
	Transverse coefficient of thermal expansion	E 831, D 696	All types
	Flash ignition temperature	D 1929	All types

Notes:

^a Where more than one test method is indicated, testing may be completed using any of the ASTM test methods indicated. All similar tests (e.g. tension, compression) must be completed using the same test method and the method used must be reported along with the test data. For tests on rods, only one of the listed test methods is appropriate.

^b Procedure A does not require the testing and reporting of flexural properties. Flexural properties (moduli and strength) may be useful for design of plate-like structures or rods that receive out-of-plane loading. Appropriate tests are ASTM D 790 for plates and D 4476 for rods. If flexural testing is used in Procedure B (see Section A.8.6), then flexural testing must be completed and reported as part of Procedure A testing.

^c For hybrid laminates, report volume of primary and secondary fibers.

^d Differential scanning calorimetry (ASTM E 1356) and dynamic mechanical analysis (ASTM E 1640) provide a direct measure of the glass transition temperature, denoted T_g . The transition temperature provided by ASTM D 648 is described as the heat deflection temperature. The transition temperature provided by ASTM E 2092 is described as the heat distortion temperature. Test methods that provide a direct measure of T_g (E 1356 and E 1640) are preferred. The two alternate methods are allowed in situations where the primary methods cannot detect the transition temperature. The property limit requirements given in Tables A.8.2, A.9.1 and A.9.2 apply, regardless of the test method selected.

^e Procedure for long-term immersion as given in ASTM D 570 shall be followed except that the temperature for conditioning shall be 50 °C (122 °F).

method, or is specified as fewer than five, then a minimum of five specimens shall be tested for each mechanical and physical property reported as a requirement of Section A.8.²⁶ The size and shape of the specimens shall be as required for testing according to the standard test methods given in Table A.8.1.

²⁶ The requirement for number of coupons to be tested in Procedure A defers to the number recommended by the individual test methods, but may be no fewer than five. For the calculation of statistically significant 'characteristic' values, a larger number of samples may be needed. Guidance for determining the number of samples for statistical significance is given in ASTM E 122, standard practice for calculating sample size to estimate, with a specified tolerable error, the average for a characteristic of a lot or process. The purchaser may specify a larger number of samples for selected properties in accordance with standard E 122.

A.8.5.4. *Pre-conditioning*—Pre-conditioning of specimens for Procedure A tests shall be in accordance with ASTM D 618 (Procedure A).²⁷

A.8.5.5. *Procedure*—Test the specimens to determine each property listed in Column 1 of Table A.8.1, using one of the test methods stipulated in Column 2 of Table A.8.1. Determine the mean property value and the coefficient of variation. Retain the individual test values

²⁷ The pre-conditioning required for Procedure A is not guaranteed to bring the FRPs into moisture equilibrium at standard laboratory conditions but should assist in achieving reproducible results for materials that may be shipped and stored in non-laboratory conditions. Achieving moisture equilibrium, especially for thick FRPs, may require excessive time periods. The pre-conditioning regime required here takes precedence over any pre-conditioning regime required by the specified test methods listed in Table A.8.1.

Table A.8.2

Property requirements for Procedure A

Material property	FRP material classification ^a					
	Glass Type 1	Carbon Type 1	Glass Type 2	Glass Type 3	Glass Type 4	Carbon Type 4
Strength properties						
Longitudinal tensile strength	≥ 655 MPa (95 ksi)	≥ 655 MPa (95 ksi)	≥ 345 MPa (50 ksi)	≥ 207 MPa (30 ksi)	≥ 414 MPa (60 ksi)	≥ 414 MPa (60 ksi)
Transverse tensile strength	NA	NA	≥ 41 MPa (6 ksi)	≥ 62 MPa (9 ksi)	NA	NA
Longitudinal compressive strength	≥ 448 MPa (65 ksi)	≥ 448 MPa (65 ksi)	≥ 241 MPa (35 ksi)	≥ 138 MPa (20 ksi)	≥ 276 MPa (40 ksi)	≥ 276 MPa (40 ksi)
Longitudinal short beam shear strength	≥ 41 MPa (6 ksi)	≥ 41 MPa (6 ksi)	≥ 31 MPa (4.5 ksi)	≥ 31 MPa (4.5 ksi)	≥ 31 MPa (4.5 ksi)	≥ 31 MPa (4.5 ksi)
Stiffness properties						
Longitudinal tensile modulus	≥ 35 GPa (5 Msi)	≥ 83 GPa (12 Msi)	≥ 21 GPa (3 Msi)	≥ 14 GPa (2 Msi)	≥ 21 GPa (3 Msi)	≥ 62 GPa (9 Msi)
Physical properties						
Fiber volume fraction ^b	≥ 50%	≥ 50%	≥ 40%	≥ 40%	≥ 30%	≥ 30%
Glass transition temperature	≥ 93.3°C (200°F)	≥ 93.3°C (200°F)	≥ 93.3°C (200°F)	≥ 93.3°C (200°F)	≥ 93.3°C (200°F)	≥ 93.3°C (200°F)
Barcol hardness	≥ 50	≥ 50	≥ 50	≥ 50	≥ 50	≥ 50
Water absorbed when substantially saturated ^c	≤ 1.0%	≤ 1.0%	≤ 2.0%	≤ 2.0%	≤ 2.0%	≤ 2.0%

Notes:

^a These provisions are based on the fiber type and laminate type. They apply to all resin types permitted by the specification. For hybrid laminates, the provisions for the primary fiber type apply.

^b For hybrid laminates the volume fraction requirement is for the combined volume of the two fibers.

^c Procedure for long-term immersion as given in D 570 shall be followed except that the temperature for conditioning shall be 50 °C (122 °F).

and supply these individual test results if required by the purchaser.

A.8.5.6. Compare the selected results from Procedure A testing with the property requirements listed in Table A.8.2. For a laminate to meet this specification, the mean property values must satisfy each inequality in Table A.8.2.

A.8.6. Procedure B—long-term material properties for qualification:²⁸ A.8.6.1. *Scope*—Procedure B uses the Arrhenius temperature dependence relationship to estimate the long-term degradation of the material at 23 °C (73 °F) through the use of heat and moisture. The

²⁸ Procedure B is based on ASTM D 3045, standard practice for heat aging of plastics without load. See also ASTM D 4502 standard test method for heat and moisture resistance of wood-adhesive joints. The philosophy of service life prediction is described in ASTM E 632, standard practice for developing accelerated tests to aid prediction of the service life of building components and materials. ASTM D 3045 gives procedures for comparing the thermal aging characteristics of materials at a single temperature and for determining the thermal aging characteristics of a single material at a series of temperatures for the purpose of estimating time to a defined property change at some lower temperature. The latter of these two procedures is the basis for Procedure B. At this time, no limits on the property retention as a function of this conditioning are specified. For specific FRP parts, it may be possible to identify target strength retentions for a given service life. If so, these provisions may be specified as special provisions that are tied to the results from Procedure B testing.

procedure provides a predicted service life for the FRP composite material at a desired level of strength retention, or provides the level of strength retained at a desired service life.

A.8.6.2. *Significance and use*—Service-life estimates based on the Arrhenius procedure rely on the assumption that the chemical degradation mechanism is the same at the elevated aging temperatures as at the service temperature, and that a single mechanism controls the degradation process throughout. The data obtained from this practice does not account for the influence of stress or other environmental factors²⁹ that exist in many actual service conditions. The method outlined below does not permit the calculation of a predicted service life if the observed degradation rates do not fit the Arrhenius assumptions.

A.8.6.3. *Summary of method*—Procedure B is based on the Arrhenius concept³⁰ of a distinct activation energy that causes long-term chemical degradation in materials. According to the Arrhenius concept, this degradation can be accelerated by subjecting the material to elevated temperature and measuring the change in material prop-

²⁹ Such as exposure to chemical environments, alkalinity, salts or UV radiation.

³⁰ Nelson, Wayne, Accelerated testing: Statistical models, Test plans and Data analyses. John Wiley and Sons, New York, 1990.

erty value as a function of conditioning time. Specimens are subjected to four elevated temperatures while immersed in deionized water for durations of 28, 56, 112 and 224 days. Specimens are tested at each of these durations for tensile or flexural strength and modulus and short beam shear strength in accordance with the appropriate test methods given in Table A.8.1. The elevated temperatures are a function of the glass transition temperature of the FRP composite. Unaged properties are determined from Procedure A by tests on specimens at time=0. The rate of property loss at each temperature can be estimated by a regression equation. This relationship may then be extrapolated to lower (service) temperatures through the use of the Arrhenius temperature dependence relationship and used to estimate the levels of property retention at the chosen service life time, or the time to reach the chosen level of property retention.

A.8.6.4. Apparatus:

A.8.6.4.1. *Water baths*—Constant-level water baths capable of control to within 0.5 °C (0.9 °F) of the desired temperature are required.

A.8.6.4.2. *Containers*—Alternatively, high density polyethylene (HDPE) containers may be filled with water. Each of these bottles must be large enough to contain a specimen. These bottles may then be placed in an oven at the required temperatures.

A.8.6.5. *Test specimens*—The number and type of the test specimens required shall be in accordance with Section A.8.5.3. Specimens shall be cut into coupons and then conditioned (individually), and cut edges shall not be sealed.

A.8.6.6. *Pre-conditioning*—Pre-condition all specimens in accordance with ASTM D 618, Procedure A.

A.8.6.7. *Specimen aging*—Four sets of specimens shall be conditioned for each type of test, each at a different elevated temperature.³¹

The temperatures are based on a reference conditioning temperature (T_{ref}), taken as 80% of the glass transition temperature as measured in Procedure A. The four temperatures for conditioning are determined as follows: The difference between the T_{ref} and 40 °C (104 °F) is divided by three to obtain the temperature interval, $\Delta T = [T_{\text{ref}} - 40 \text{ °C}] / 3$ ($\Delta T = [T_{\text{ref}} - 104 \text{ °F}] / 3$). The four conditioning temperatures are $T_1 = T_{\text{ref}}$, $T_2 = T_{\text{ref}} - \Delta T$, $T_3 = T_{\text{ref}} - 2\Delta T$ and $T_4 = T_{\text{ref}} - 3\Delta T$. Specimens are immersed in de-ionized water at each of the specified temperatures for periods of 28, 56, 112 and 224 days.³²

³¹ ASTM D 3045 requires testing at a minimum of four elevated temperatures. The more temperatures used, the more data is available to make a prediction of property retention at the service life time. Theoretically, this will lead to a more accurate prediction. ASTM D 4502 requires 5 temperatures.

³² Appropriate test results from Procedure A shall be used in lieu of testing specimens at time=0 (t_0).

A.8.6.8. Procedure

A.8.6.8.1. Obtain the results of the appropriate ASTM standard tests for longitudinal tensile or flexural strength and modulus and short beam shear strength as required in Table A.8.1.

A.8.6.8.2. Age additional sets of specimens for periods of 28, 56, 112 and 224 days at each of the four temperatures selected according to the method presented in Section A.8.6.7. At the end of each aging period, remove specimens from the conditioning environment and test at room temperature without allowing the specimens to dry out. Test one set of specimens aged at each temperature for tensile (or flexural) strength and modulus, and one set for short beam shear strength, in accordance with the appropriate ASTM test methods (see Table A.8.1).

A.8.6.8.3. Record the time of exposure and test results for each test.

A.8.6.9. *Calculation*—Calculate the average property value for specimens tested at each temperature. Calculate the property retention (PR) values for each property as the average property value at the time of testing (t_i) divided by the average property value at time=0 (t_0). Plot these data on a chart similar to Fig. A1a and b, with time on the horizontal axis using a logarithmic scale, and the property retention value on the vertical axis using a linear scale. Using linear regression, fit a line through each set of data (one for each aging temperature). An acceptable regression line must have an r^2 of at least 0.80. If one or more of the regression lines does not have an acceptable r^2 , then the Arrhenius plots as described in Section A.8.6.9.1 through A.8.6.9.4 should not be calculated. The property retention data must still be reported.

A.8.6.9.1. The Arrhenius plot may be constructed in one of two ways. First, time may be plotted as a function of inverse absolute temperature for various percentages of property retention. Alternatively, property retention may be plotted as a function of inverse absolute temperature for various chosen lifetimes.

A.8.6.9.2. Determine the time to reach given levels of property retention at each of the conditioning temperatures. This is done by substituting various values of property retention into the regression equations plotted on Fig. A1a. Graphically, this is represented as drawing horizontal lines on Fig. A1a at, for example, 80, 70, 60 and 50% retention, and then drawing vertical lines from the points where these lines intersect the regression lines for the various temperatures.³³ These vertical lines represent the times at which the various retention percent-

³³ This is shown in Fig. A1a for the 60% retention property value.

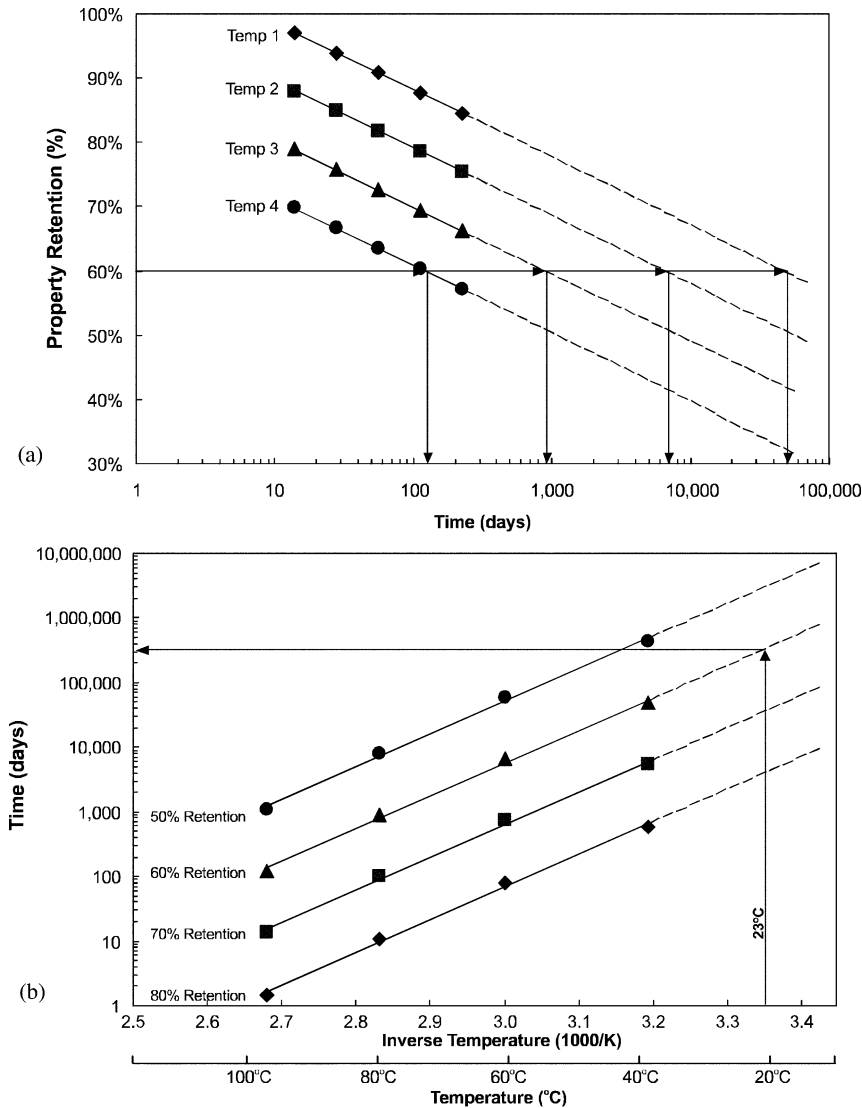


Fig. A1. (a) Plot of property retention as a function of time and (b) Arrhenius plot for service life as a function of temperature and percent retention.

ages occur. For each chosen level of property retention, plot the time (days) on a logarithmic scale as a function of the reciprocal of the absolute temperature ($1000/^\circ\text{K}$), as in Fig. A1b (Arrhenius plot). Use regression analysis to determine the equations, defining the relationships between the time and the reciprocal of the absolute temperature for each level of property retention. Extrapolate these relationships (represented by dashed lines in Fig. A1b) to determine the time to reach the chosen levels of property retention at 23°C (73°F).

A.8.6.9.3. Alternatively, time values, representing service life times, may be substituted into the regression equations in Fig. A2a. This is represented graphically by vertical lines drawn on Fig. A2a at lifetimes such as

10, 25, 50 and 75 years.³⁴ The values of property retention for each of the chosen lifetimes at each of the temperatures on Fig. A2a are represented by drawing horizontal lines from the intersection of the vertical lines with the various temperature regression lines. For each chosen lifetime, plot the property retention (%) as a function of the inverse temperature ($1000/^\circ\text{K}$), as in Fig. A2b. Use regression analysis to determine the equations defining the relationships between the property retention and the reciprocal of the absolute temperature for each service lifetime chosen. Extrapolate these relationships (represented by dashed lines in Fig. A2b) to

³⁴ This is shown in Fig. A2a for the 10 year (3650 day) service life value.

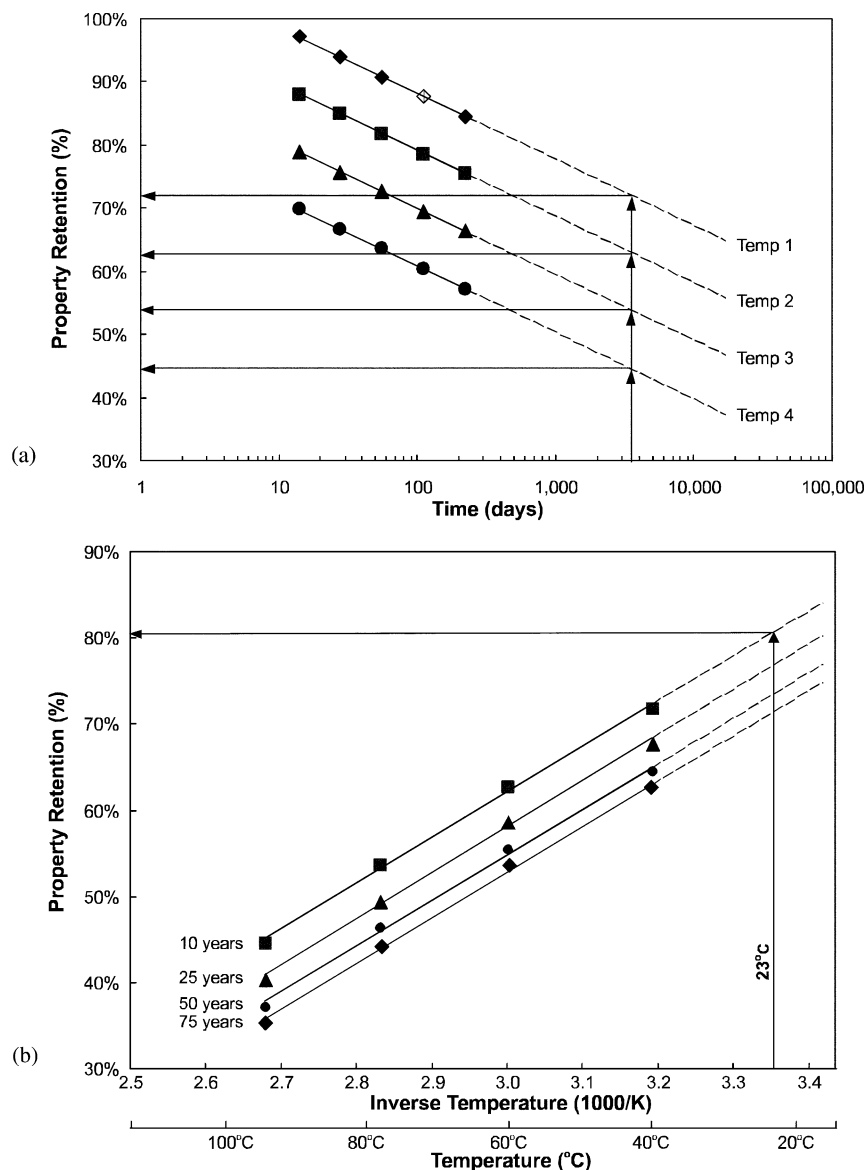


Fig. A2. (a) Plot of property retention as a function of time and (b) Arrhenius plot for property retention as a function of temperature and service life.

determine the levels of property retention at the chosen life times at 23 °C (73 °F).

A.8.6.9.4. Calculate the 95% confidence limit³⁵ for the predicted value(s) from a regression analysis for the predicted lifetime or the estimated property retention.

A.8.6.10. *Optional environments for procedure B*—The de-ionized water solution called for in this procedure may be replaced by an alkaline or salt solution as

³⁵ The 95% confidence limit can be taken as a conservative estimate of the predicted value of the property under investigation. However, as the accuracy and precision of this method have not been characterized, the usefulness of the 95% confidence limit may be limited in this case. See Annex B2 of ASTM D 4502, standard test method for heat and moisture resistance of wood-adhesive joints, for example calculations of the 95% confidence limit.

described below. De-ionized water is the preferred solution.

A.8.6.10.1. *Alkaline solution*—Dissolve 1185-g $\text{Ca}(\text{OH})_2$, 9-g NaOH and 42-g KOH in 10-l water.

A.8.6.10.2. *De-Icing salt solution*—Dissolve 5 ± 1 parts by mass of sodium chloride in 95 parts water, with a resultant pH of the solution between 6.5 and 7.2 as per ASTM B 117 Standard Practice for Operating Salt Spray (Fog) Apparatus.

A.8.6.10.3. *Substitute sea water*—Prepare a substitute sea water solution per ASTM D 1141 Standard Practice for the Preparation of Substitute Ocean Water.

A.8.7. *Optional long-term properties*. A.8.7.1. *Scope*—Conditioning and testing of specimens in other environ-

Table A.9.1

Material requirements for Procedure C (Maximum permissible deviations from qualification values determined in Procedure A)

Material property	FRP material classification ^a					
	Glass Type 1 (%)	Carbon Type 1 (%)	Glass Type 2 (%)	Glass Type 3 (%)	Glass Type 4 (%)	Carbon Type 4 (%)
Strength properties						
Longitudinal tensile strength	± 10	± 10	± 10	± 10	± 10	± 10
Longitudinal compressive strength	± 10	± 10	± 10	± 10	± 10	± 10
Longitudinal short beam shear strength	± 10	± 10	± 10	± 10	± 10	± 10
Stiffness properties						
Longitudinal tensile modulus	± 5	± 5	± 5	± 5	± 5	± 5
Physical properties						
Fiber volume fraction	± 5	± 5	± 5	± 5	± 5	± 5
Thickness	± 5	± 5	± 5	± 5	± 5	± 5
Glass transition temperature	± 5	± 5	± 5	± 5	± 5	± 5
Barcol hardness	± 5	± 5	± 5	± 5	± 5	± 5
Water absorbed when substantially saturated ^b	± 5	± 5	± 5	± 5	± 5	± 5

Notes:

^a These provisions are based on the fiber type and laminate type. They apply to all resin types permitted by the specification. For hybrid laminates, the provisions for the primary fiber type apply.

^b Procedure for long-term immersion as given in D 570 shall be followed except that the temperature for conditioning shall be 50 °C (122 °F).

ments may be deemed appropriate as qualification procedures for FRP composite laminates.³⁶ These tests may be performed as agreed upon between purchaser and manufacturer.

A.8.7.2. Creep rupture strength—Creep rupture strength refers to the stress that will cause fracture in a sustained load test at a given time. This test provides data on the behavior of a material under constant load. ASTM test method D 2990 standard test methods for tensile, compressive and flexural creep is recommended as the method to characterize this phenomenon.

A.8.7.3. Fatigue life—Fatigue life is a measure of the number of cycles of stress or strain that a specimen can sustain without failing. The recommended test method for fatigue life is ASTM D 3479 standard test method for tension–tension fatigue of polymer matrix composite materials.

A.8.7.4. Freeze-thaw resistance—Freeze thaw conditioning consists of alternately raising and lowering the temperature of specimens above and below 0 °C (32 °F), according to ASTM C 666 standard test method for resistance of concrete to rapid freezing and thawing. Mechanical tests from Table A.8.1 can be selected to characterize the retention of mechanical and physical properties after freeze-thaw conditioning. These values

may then be compared to the values obtained in Procedure A testing.

A.9. Physical and mechanical testing for part acceptance

A.9.1. Scope. Any given FRP composite part made according to this specification must be produced from one or more qualified laminates that have met the requirements of Section A.8. Section A.9 presents sampling and testing procedures for verifying that coupons cut from the actual part give substantially equivalent test results to those obtained during qualification testing.³⁷ Two testing procedures are required for the coupons cut from the actual part. Procedure C (short-term material properties for acceptance) provides procedures for testing specified mechanical and physical properties of coupons cut from the part and compares these to the test data from the qualified laminate on which the part is based (using Procedure A data). Procedure D (short-term material properties for acceptance in the saturated state) determines specified mechanical and physical properties of coupons in the saturated condition, to demonstrate that mechanical and

³⁶ These optional long-term properties are not included as coupon-level material testing. These tests are more appropriate for FRP parts or full-section FRP assemblies. However, situations may arise where a purchaser wishes to have these tests conducted on the material to be qualified under this specification. Thus, these optional sections may be required by some purchasers.

³⁷ The specification does not dictate the frequency at which FRP composite parts should be sampled and tested according to the acceptance procedure described in Section 9. Manufacturers must provide certifications of their products and must have adequate quality control procedures to ensure that all lots of their materials meet the specification. Optional, independent verification of the acceptance testing described in Section 9 can be specified by invoking Section 13 (Quality Assurance) of this specification.

Table A.9.2

Material requirements for Procedure D—retained properties in the saturated state

Material property	FRP Composite material classification ^{a,b}					
	Glass Type 1	Carbon Type 1	Glass Type 2	Glass Type 3	Glass Type 4	Carbon Type 4
Strength properties						
Retained longitudinal tensile strength	≥ 95%	≥ 95%	≥ 90%	≥ 90%	≥ 90%	≥ 90%
Retained longitudinal compressive strength	≥ 95%	≥ 95%	≥ 90%	≥ 90%	≥ 90%	≥ 90%
Retained Longitudinal short beam shear strength	≥ 95%	≥ 95%	≥ 90%	≥ 90%	≥ 90%	≥ 90%
Stiffness properties						
Retained longitudinal tensile modulus	≥ 95%	≥ 95%	≥ 95%	≥ 95%	≥ 95%	≥ 95%
Physical properties						
Glass transition temperature ^c	≥ 71.1 °C (160 °F)	≥ 71.1 °C (160 °F)	≥ 71.1 °C (160 °F)	≥ 71.1 °C (160 °F)	≥ 71.1 °C (160 °F)	≥ 71.1 °C (160 °F)

Notes:

^a Minimum requirements for mechanical strength are given as percentage retention from values observed in Procedure C testing. Where more than one test method is applicable, the test method selected in Procedure D shall be the same as that selected for Procedure C.

^b These provisions are based on the fiber type and laminate type. They apply to all resin types permitted by the specification. For hybrid laminates, the provisions for the primary fiber type apply.

^c Wet glass transition temperatures are recorded as the ‘first scan’ transition temperatures. If second scan temperatures are completed, T_g at the second scan should also be reported.

physical properties experience little property loss in the saturated moisture condition at 50 °C (122 °F).

A.9.2. Procedure C—short-term material properties for acceptance. A.9.2.1. *Scope*—Each distinct laminate within the part is sampled and cut into coupons as specified in Section 9.4. These coupons are subjected to mechanical and physical testing as described in Table A.9.1. Results from these tests are compared with the results from tests completed on the qualified laminate (see Table A.8.1). Materials are accepted according to the specification if reported values are within the maximum deviation given in Table A.9.1. The same test methods used on the qualification laminates must be used on coupons cut from the part.

A.9.2.2. *Significance and use*—In order for a part to be accepted, the test values from Procedure A testing and Procedure C testing must agree within the range specified in Table A.9.1.

A.9.2.3. *Test specimens*—The number and type of specimens and sampling locations shall be in accordance with Section A.9. The size and shape of specimens shall be as required for testing according to the standard test methods given in Table A.8.1.

A.9.2.4. *Pre-conditioning*—Pre-conditioning of specimens shall be in accordance with ASTM D 618 (Procedure A).

A.9.2.5. *Procedure*—Test the specimens to determine each property listed in Column 1 of Table A.9.1, using the standard test methods specified in Table A.8.1. Determine the mean property value and coefficient of

variation. Retain these individual test values and supply these individual test results if required by the purchaser.

A.9.2.6. Compare the values obtained in Section A.9.2.5 to those obtained in Section A.8.5 (Procedure A). Determine the percent of variation of the mean of Procedure C test values from the mean of the same tests performed in Section A.8.5.5. The two sets of values shall agree within the range specified in Table A.9.1.

A.9.3. Procedure D—short-term material properties for acceptance in the saturated state. A.9.3.1. *Scope*—Procedure D provides a method for moisture conditioning and subsequent testing to provide mechanical property data for the FRP composite in a saturated condition.³⁸

³⁸ The limiting residual property values determined from testing in Procedure D and given in Table A.9.2 should not be used in design codes as property values for structural design. The residual property values obtained in Procedure D are not intended as an indication of the long-term durability of the material, nor should they be assumed to be the long-term property values for the material (Procedure B does provide this information). Procedure D conditioning is intended to provide an indication of whether or not the FRP material is sufficiently robust to retain a significant percentage of its as-produced properties following exposure to a short-duration moderately-elevated temperature in the saturated state. The conditioning is intended to simulate a possible, although extreme, short-term service operating environment for the material. Both elevated temperature and moisture are known to reduce the mechanical properties of FRP composite materials. However, the Procedure D conditioning is not intended to be an indicator of material degradation. The changes in material properties due to the presence of absorbed water when the material is conditioned according to Procedure D are assumed to be reversible, as opposed to the changes observed during Procedure B testing, which are assumed to be irreversible.

A.9.3.2. *Significance and use*—Mechanical properties and the glass transition temperature of an FRP composite may be reduced when the FRP composite is in a saturated moisture condition. Field conditions for FRP composites in highway applications will be less favorable than laboratory conditions and may approach that of a saturated state for some situations. Therefore, the reporting of mechanical test data in a saturated state is required. Additionally, it has been demonstrated that the properties of high quality FRP composites constructed of compatible fiber and resin systems are not significantly affected by moisture. Therefore, limits on the property changes in saturated moisture conditions are given in Table A.9.2.

A.9.3.3. *Pre-conditioning*—Immerse test specimens for tests described in Table A.9.2 in water at 50 °C (122 °F) until the specimens reach their saturated state as determined in Procedure A by ASTM D 570—procedure for long term immersion.

A.9.3.4. *Procedure*—Test conditioned specimens to determine the properties in the saturated state using the same test methods used in Procedure C. Compare these saturated properties with the limits given in Table A.9.2. The values determined in Procedure D shall meet or exceed the percent retention values given in Table A.9.2.

A.9.4. *Sampling requirements for acceptance testing.*

A.9.4.1. *Scope*—In this section, requirements for cutting coupons for subsequent mechanical and physical testing are provided. Definitions of the longitudinal and transverse directions for the laminate are provided. For laminates that cure on and are bonded to another material, a method for the production of a ‘witness laminate’ is given.

A.9.4.2. *Coupon sampling*—Each distinct laminate within the part shall be sampled. Mechanical and physical test coupons of the appropriate size for the test method selected for longitudinal mechanical and physical testing listed in Table A.9.1 shall be cut from the part.³⁹ Coupons shall be selected without bias, that is, the coupons shall be selected from all regions of the part that are constructed of the distinct laminate. Annex B2 gives an example part and demonstrates the required samples for two of the distinct laminates within the part.

A.9.4.2.1. *Longitudinal and transverse directions*—Each part that is tested according to Section A.9 shall have a designated longitudinal direction. The manufacturer shall select the longitudinal direction for sampling and mechanical testing of coupons cut from this part. The manufacturer must designate either the primary structural direction for the part or the primary direction of production for the part as the longitudinal

direction.⁴⁰ The longitudinal direction for all laminates within a part must be parallel to the part longitudinal direction. For a given laminate, the transverse direction is always perpendicular to the longitudinal direction and in the plane of the laminate.⁴¹ The longitudinal axis of the test property laminates described in Section A.8 must match the longitudinal axis of the laminates cut from the actual parts.

A.9.4.2.2. The number and type of samples shall be in accordance with the ASTM test method for the specific property to be determined (see Table A.8.1). However, if the number of specimens is not specified in the test method, or is specified as fewer than five, then a minimum of five specimens shall be tested for each mechanical and physical property reported as a requirement of Section A.9.

A.9.4.2.3. Coupons shall not be cut through junctions in complex parts. In addition, if the part contains thickened regions in the vicinity of junctions, then the coupons must not include material from these thickened regions.

A.9.4.3. *Witness laminates*—If the FRP composite material is produced on a substrate or field-cured onto another material that prevents it being tested in a stand-alone fashion, then a witness laminate must be made to characterize the mechanical and physical properties of the material.⁴² Witness laminates shall be produced contemporaneously in the field, under the same environmental and cure conditions, and using the same lay-up and laminate consolidation processes as the laminates produced on the substrate.⁴³

⁴⁰ In many cases, the “primary structural direction” and the “primary direction of production” will coincide. Some FRP production methods such as pultrusion have a definite production axis that is typically aligned with the primary load-carrying axis of the part. Other open- and closed-molded production processes have no definite axis of production. In this case, the longitudinal axis must be defined by the primary structural direction (i.e. the direction of maximum axial stress). This may or may not coincide with the principal production direction.

⁴¹ A part has one unique longitudinal direction, since the longitudinal direction for all distinct laminates in the part is parallel to the part’s longitudinal direction. The part does not have a single unique transverse direction. However, each laminate does have a single, unique transverse direction.

⁴² For field-cured FRPs used to repair concrete structures in the field, it is customary to derive mechanical and physical properties of the FRP from witness laminates (sometimes called witness panels). This is distinct from the TPL described in Section A.8.3. A witness panel is constructed in the field contemporaneously with the actual construction, and tested afterwards. A TPL is manufactured and tested in the plant before construction.

⁴³ Additional structural-level tests of the FRP composite material and its substrate may be required to validate the performance of the FRP composite and to characterize the bond between the FRP composite and its substrate. Requirements for such tests are beyond the scope of this specification.

³⁹ Some Class 1 laminates may be suitable for testing without cutting into coupons.

A.10. Workmanship, finish and appearance

A.10.1. The finished material shall be smooth, clean, uniform in color and reasonably free from conducting particles, foreign materials, pits, cracks, voids, chips, sink marks, delaminations, blisters and scratches, in accordance with good commercial practice. The distribution of filler, additives and fibers shall be in accordance with good commercial practice. The material shall be free of resin-rich and resin-starved areas, and there shall be no evidence of significant reinforcement shifting, wrinkles, bunching or density variation within a length, all in accordance with good commercial practice.⁴⁴

A.11. Certification and test report

A.11.1. *Certification.* The manufacturer's certification shall be furnished to the purchaser stating that each lot of material has been manufactured in accordance with this specification and that the requirements herein have been met.

A.11.2. *Test report.* For each distinct laminate within a part, the test report shall contain the information described in this section. An authorized agent of the manufacturer shall sign, date and certify all test reports. The manufacturer shall provide two copies of the certified test report at the time of material delivery.

A.11.2.1. *Classification*—Report the laminate identification code as described in Section A.8.2.

A.11.2.2. Materials

A.11.2.2.1. Report type and manufacturer of resin.

A.11.2.2.2. Report manufacturer and amount of styrene included as an integral part of the resin (from the resin supplier).

A.11.2.2.3. Report primary fiber type and manufacturer.

A.11.2.2.4. Report secondary fiber type and manufacturer, if any.

A.11.2.2.5. Report amount by volume of secondary fiber, if any.

A.11.2.2.6. Report fiber architecture type (Type 1, 2, 3 or 4, as described in Section A.4.5).⁴⁵

A.11.2.2.7. Report type and manufacturer of filler, diluent (styrene), and low-profile additives, if any, and amount as a percentage of the weight of the base resin.

A.11.2.2.8. Report type and manufacturer of fiber treatments, coupling agents, release agents, initiators, hardeners, promoters, catalysts, UV agents, fire retardants, foaming agents, gelcoats, surfacing treatments/veils and pigments.

A.11.2.3. Production:

A.11.2.4. Report type of manufacturing process used (e.g. pultrusion, vacuum molding, hand lay-up).

A.11.2.5. Report temperature of die or mold, or ambient temperature during manufacturing process, and duration and temperature of post-curing (if used).

A.11.2.6. Report the definition of a production lot.

A.11.2.7. Report the average value and range for the amount of material in a production lot.

A.11.2.8. Report date of production for each lot of material.

A.11.2.9. *Material characterization—Procedure A*—Report a complete set of test results from Section A.8, Procedure A, as given in Table A.8.1 (including flexural strength and modulus tests, if completed).

A.11.2.9.1. Report test methods used, number of samples tested, mean⁴⁶ and coefficient of variation for each property measured.

A.11.2.9.2. Report any deviations from the standard test method.

A.11.2.9.3. Report whether the FRP composite laminate tested meets the specification requirements for each property measured, as given in Table A.8.2.

A.11.2.10. *Long-term material characterization—Procedure B*

A.11.2.10.1. Report manufacturer and model of aging oven used.

A.11.2.10.2. Report type of aqueous solution used.

A.11.2.10.3. Report aging temperatures: T_1 , T_2 , T_3 and T_4 .

A.11.2.10.4. Report test method used to determine tensile strength and modulus.

A.11.2.10.5. Report test method used to determine short beam shear strength.

A.11.2.10.6. Report any deviations from the test methods used.

A.11.2.10.7. Report visible changes in conditioned test specimens, if any, before testing.

A.11.2.10.8. For each set of specimens, report the aging temperature, actual aging time, mean property value,⁴⁷ number of samples tested, coefficient of variation and property retention (as a percentage of Procedure A values).

A.11.2.10.9. For each property measured (i.e. tensile strength, tensile modulus and short beam shear strength), submit a plot of the property value as a function of time

⁴⁴ 'American National Standard for Ladders – Portable Reinforced Plastic.' American National Standards Institute (ANSI) A14.5-2000.

⁴⁵ The manufacturer is required to report the type of fibers and fabrics in the laminates. Similarly, the type and manufacturer of the resin, diluents and additives must be disclosed. The weight percent of each additive need not be provided unless noted otherwise. This specification does not require that the exact stacking sequence and number of layers of various fabrics or number of rovings or tows within the laminate be provided. The purchaser of the materials or parts may require that the stacking sequence be provided. If so, this information should be requested by the purchaser at the time of ordering.

⁴⁶ Manufacturer shall retain individual test results, and shall provide this data if requested by purchaser.

⁴⁷ Manufacturer shall retain individual measurements and shall provide this data if requested by purchaser.

at each temperature measured (as in Fig. A1a and Fig. A2a).

A.11.2.10.10. For each aging temperature, report the regression equation developed for property changes as a function of exposure time,⁴⁸ as in Fig. A1a and/or Fig. A2a, as well as the associated r^2 value.

A.11.2.10.11. For each level of property retention, report the time required to reach the given level of property retention at each temperature.⁴⁹

A.11.2.10.12. Submit a plot of time as a function of inverse temperature for each level of property retention, as in Fig. A1b.

A.11.2.10.13. For each level of property retention, report the regression equation developed for lifetime as a function of inverse temperature, as in Fig. A1b.

A.11.2.10.14. For each lifetime, report the property retention level at each temperature.

A.11.2.10.15. Submit a plot of property retention as a function of inverse temperature for each service lifetime, as in Fig. A2b.

A.11.2.10.16. For each level of property retention, report the regression equation developed for property retention as a function of inverse temperature, as in Fig. A2b.

A.11.2.10.17. Report residual properties and/or service lifetimes at 23 °C (73 °F) for each property (i.e. tensile strength, tensile modulus and short beam shear strength).

A.11.2.10.18. Report 95% confidence limit for predicted values at 23 °C (73 °F) for each property (i.e. tensile strength, tensile modulus and short beam shear strength).

A.11.2.11. *Other long-term tests:*

A.11.2.12. Report other optional long-term tests conducted, if any, and results as required in specified test method(s).

A.11.2.13. *Material acceptance—Procedure C*—Report a complete set of test results from Section A.9.2, Procedure C, as given in Table A.9.1.

A.11.2.13.1. For each mechanical and physical property, report the test method used, number of samples tested, mean of specimen results,⁵⁰ and coefficient of variation.

A.11.2.13.2. For each mechanical and physical property, report the percent of variation of the mean of Procedure C tests from the mean of the same tests reported in Section A.11.2.9.1 from Procedure A testing.

A.11.2.13.3. For each mechanical and physical property, report whether the FRP composite laminate meets the specification requirements, as given in Table A.9.1.

A.11.2.13.4. Report any deviations from the standard test methods.

A.11.2.14. *Material acceptance in the saturated state—Procedure D*—Report a complete set of test results from Section A.9.3, Procedure D, as given in Table A.9.2.

A.11.2.14.1. For each mechanical and physical property, report the test method used, number of samples tested, mean of specimen results,⁵¹ and coefficient of variation.

A.11.2.14.2. For each mechanical and physical property, report the percent of variation of the mean of Procedure D tests from the mean of the same tests reported in Section A.11.2.13.1 from Procedure C testing.

A.11.2.14.3. For each mechanical and physical property, report whether the FRP composite laminate meets the specification requirements.

A.11.2.14.4. Report any deviations from the standard test methods.

A.11.2.15. *Sampling:*

A.11.2.15.1. Submit a drawing of the part, with each distinct laminate identified along with its classification. The location of coupon extraction from the part shall also be shown in this drawing (see Annex B2 for example).

A.11.2.15.2. The manufacturer shall report the longitudinal direction for testing on this drawing.

A.11.2.15.3. Report the use of witness laminates, if applicable.

A.12. Product marking

A.12.1. Each part shall be continuously marked with the designation of this specification (with which the part complies), manufacturer's name, date of production and manufacturer's part identification number.⁵² Markings shall occur at least every 2 m (6.5 foot) in linear measurement or at least every 10 m² (105 foot²) in areal measurement.

A.13. Quality assurance

A.13.1. *Scope.* The sampling and testing outlined in this section are optional acceptance tests that are completed

⁴⁸ Regression equations must be developed and reported for each property (i.e. tensile strength, tensile modulus, and short-beam shear strength).

⁴⁹ Sections A.11.2.10.11 through A.11.2.10.18 need not be reported if the r^2 value reported in A.11.2.10.10 is not greater than or equal to 0.80.

⁵⁰ Manufacturer shall retain individual measurements and shall provide this data if requested by purchaser.

⁵¹ Manufacturer shall retain individual measurements and shall provide this data if requested by purchaser.

⁵² The specification does not dictate the format of the manufacturer's part identification number. The nomenclature for part identification is left to the manufacturer. The laminate identification number is of a standard format, as specified in Section A.8.2, and must be reported along with the test data. The laminate identification number(s) need not be placed on the production parts.

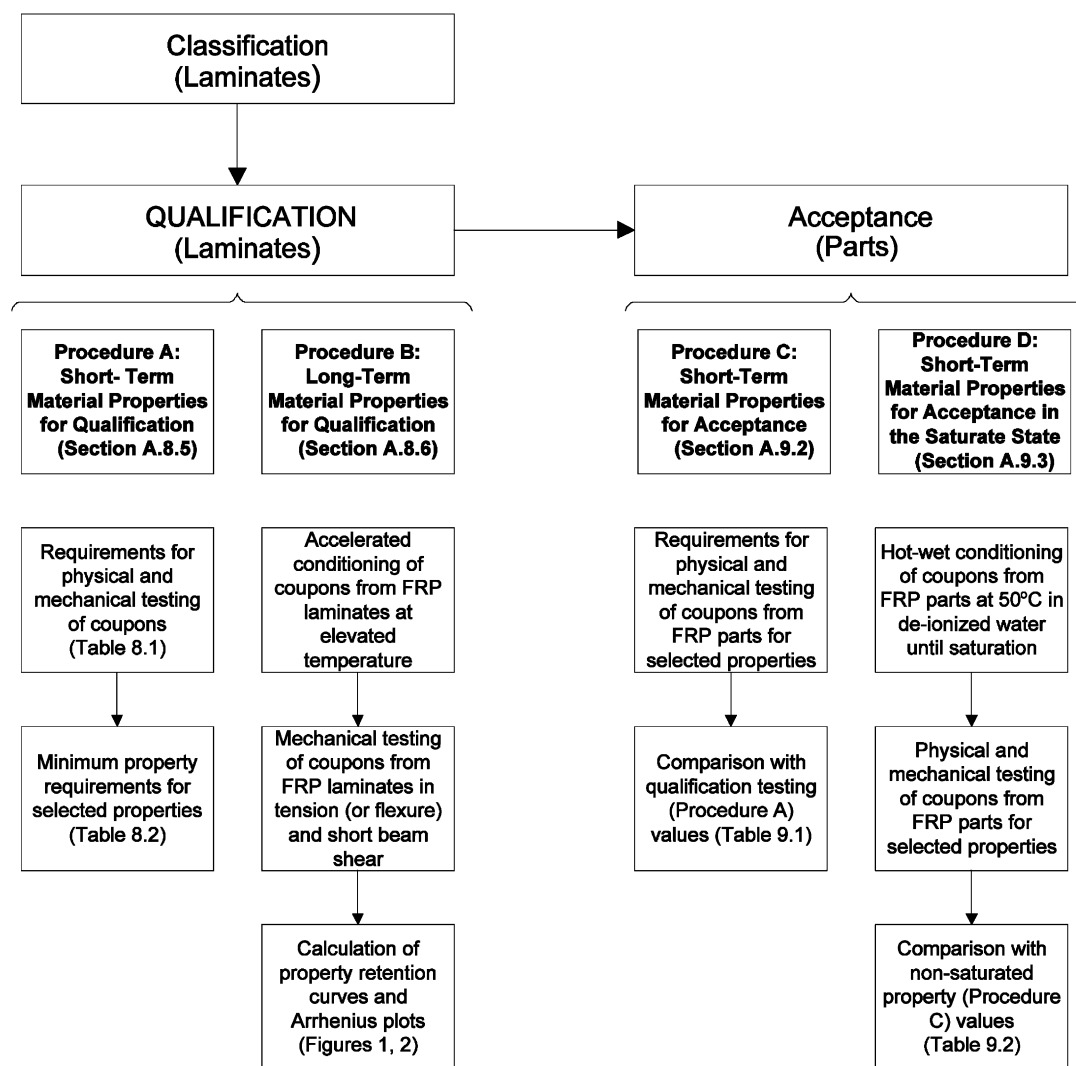


Fig. B1.1. Schematic of qualification and acceptance testing requirements.

by the purchaser or an independent laboratory⁵³ to ensure that the material provided for a given project meets the requirements of the specification and to verify that the mechanical properties of the materials supplied match those reported in the test reports (see Section A.11).

A.13.2. Sampling. Each lot of parts shall be sampled and tested. Each distinct laminate within the part shall be tested. Samples shall be taken from production parts at locations determined by the manufacturer and purchaser. Samples shall represent production occurring at the beginning, in the middle and at the end of the production run if the lot is produced as part of a continuous process.

⁵³ Quality assurance tests may also be completed by the manufacturer and provided to the purchaser as certified, notarized test reports. Such reports should note the date the materials were produced, the date they were sampled, the lot or batch number, the date the materials were tested, and the test results.

A.13.3. Testing. Mechanical and physical tests shall be completed using the same test methods and requirements specified in Section A.8 and A.9. The type and number of tests shall be as required by the purchaser. The Procedure C tests and limits outlined in Table A.9.1 may be used for quality assurance testing.

A.14. Keywords

Fiber-reinforced polymer composites, mechanical properties, physical properties, accelerated aging, material requirements, classification.

Appendix B: ANNEXES

B.1. Flowchart of qualification and acceptance testing requirements

B1.1. Fig. B1.1 depicts the overall requirements for mechanical and physical testing of coupons for material qualification (Procedures A and B) and acceptance (Procedures C and D).

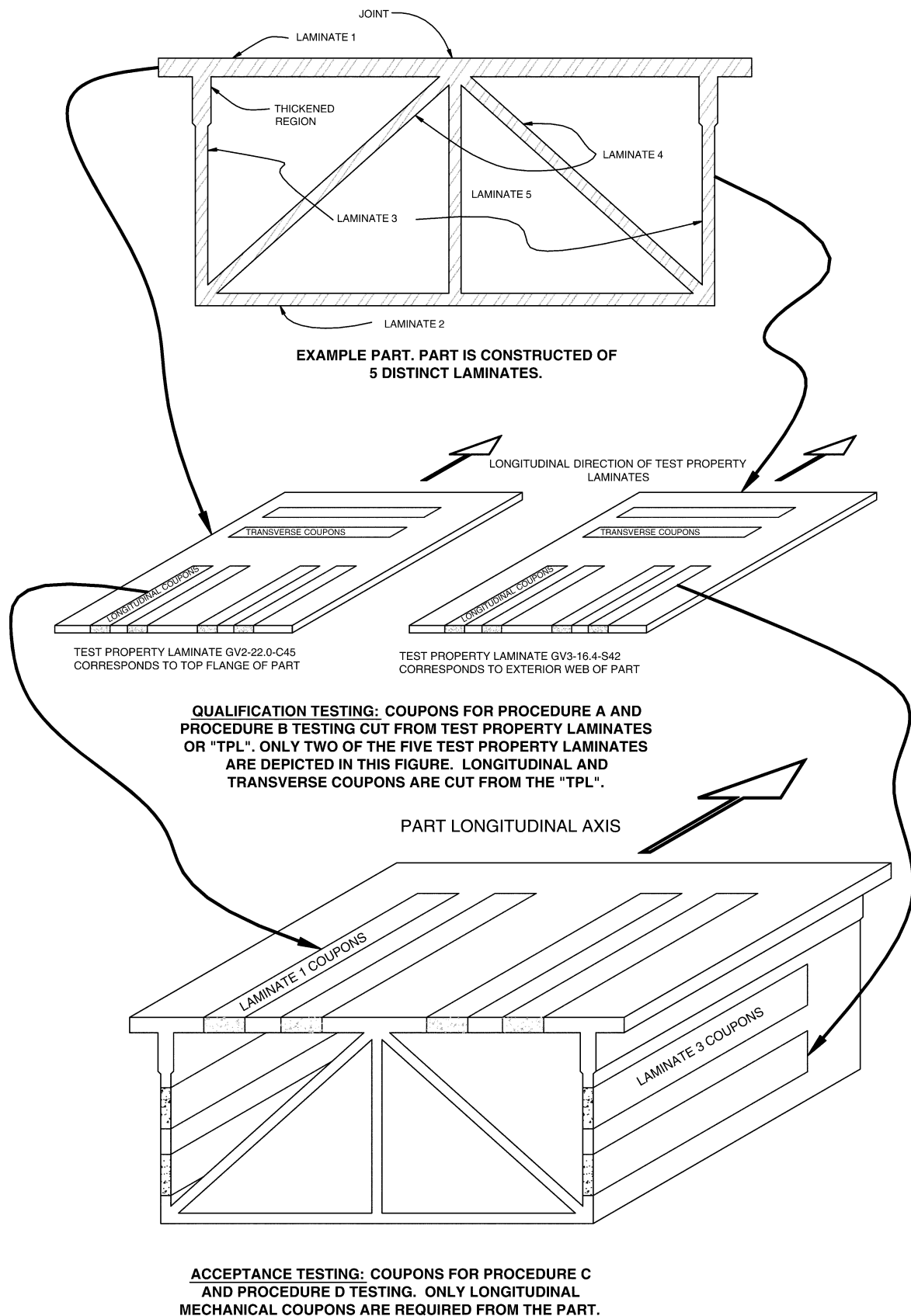


Fig. B2.1. Example part and qualification, acceptance, and sampling requirements.

Table B2.1

Example laminate identification codes

Laminate	Identification code
1	GV2-22.0-C45
2	GV2-19.4-T11
3	GV3-16.4-S42
4	GV3-12.2-S19
5	GV3-19.1-S67

B.2. Discussion of coupon sampling requirements

B2.1 Fig. B2.1 shows an FRP composite part. The part is assumed to function as a deck section—a beam with primarily flexural loading. The part's longitudinal axis coincides with the longitudinal direction of the beam. The cross-section of the part is made up of 5 distinct composite laminates, numbered 1–5 in the cross-section. The fiber architecture of each of the laminates is not detailed in this example. Each of the 5 elements in the cross-section has a unique thickness and/or load-carrying requirement and thus will have a unique fiber architecture (Table B2.1).

B2.2. *Qualification testing*—The test property laminates (TPLs) for laminates 1 and 3 are depicted in figure 3. All 5 TPLs must be produced and tested according to Section A.8 of this specification in order to qualify the laminates. In practice, it is suggested that the 5 laminates will be taken from a 'library' of laminates that the composites manufacturer has previously produced and qualified. If this is the case, then the laminates can be considered to be 'pre-qualified' and the qualification step can be skipped. If the laminates are not pre-qualified, then characterization of each TPL as described in Section A.8 must be completed.

B2.3. *Acceptance testing*—Acceptance testing completed in accordance with Section A.9 of the specification is intended to ensure that coupons cut from the part have substantially equivalent properties to those observed in the test property laminate. Coupons for acceptance testing are cut directly from production lots of the part. These acceptance-testing coupons are depicted in Fig. B2.1. Four longitudinal coupons are shown for laminate 1 and laminate 3. These four coupons represent the complete range of locations from which these laminates can be sampled. All locations within the part that are from the same distinct laminate must be sampled and pooled for subsequent mechanical testing according to Table A.9.1. This same pool of samples will provide specimens for Procedure D testing in the substantially saturated state as described in Section A.9.3 of the specification.

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